

Learning Segmentation and Economic Development*

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Abstract

How does learning from others shape economic development? We answer this question with a quantitative model of multiple labor markets and modern and traditional sectors, capturing key features of dual economies in developing countries. Workers sort across sectors and labor markets and accumulate human capital by learning from peers. Learning opportunities are sector-specific and depend on peer composition within the labor market. The model makes one main prediction: more skilled workers sort into the modern sector and, as a consequence, workers in this sector experience faster wage growth because they are exposed to higher-skilled peers. We test this prediction using longitudinal worker data from Chile and find empirical support for it. Finally, we estimate the model and use it to quantify how income per capita and welfare would change if traditional-sector workers had access to the same learning opportunities as modern-sector workers.

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1 Introduction

Human interactions are a crucial source of learning. People learn from others in the workplace, in the places they live, or from individuals in similar occupations who perform similar tasks. Lower-income economies tend to be characterized by dual economies, in which workers are split between modern and traditional sectors that coexist despite large differences in scale. This idea, central to the study of economic development, goes back to [Lewis \(1954\)](#) and is often reflected in agricultural versus non-agricultural productivity gaps, rural versus urban income differences, and the coexistence of many small firms alongside relatively few large firms within the same industries.¹ In turn, such dual economies likely shape who interacts with whom, segmenting the learning opportunities available to workers. In this paper, we study how these segmented learning markets influence workers’ skill formation and, through this channel, economic development. How do segmented learning markets affect workers’ learning? How does workers’ human capital influence their labor market choices? What are the aggregate implications for income and welfare?

We answer these questions using a quantitative model of learning from peers in a dual economy with segmented interactions, together with panel data on workers from Chile. The model features multiple labor markets and include modern and traditional sectors. Workers differ in skill, choose where to work, and accumulate human capital by learning from peers in their labor market. Importantly, learning is segmented within labor markets, with a worker’s learning depending on the skill composition of peers in their own sector. The model predicts that higher-skill workers select into the modern sector. It also predicts a dynamic modern-sector wage premium, with modern-sector workers experiencing faster wage growth when learning is sufficiently segmented. We then test these predictions using the Chilean panel, which allows us to track workers across labor markets and across the modern and traditional sectors. Consistent with the model, modern-sector workers earn higher wages and experience faster wage growth. We provide suggestive evidence that this dynamic premium reflects learning from peers: exposure to higher-paid, more skilled peers predicts higher future wage growth, and modern-sector workers are more exposed to such peers. Finally, we estimate the model and use it to quantify how much of the low income levels in developing countries stem from lost learning opportunities for workers in the traditional sector.

The model is designed to capture key features of dual economies in developing countries. Labor markets differ in productivity and include a modern and a traditional sector. Modern-sector firms operate a scalable constant-returns technology but face hiring frictions in the form of payroll taxes. Traditional-sector firms avoid these regulations, but operate a decreasing-returns technology, capturing small-scale production or subsistence self-employment. Workers differ in skill, choose where to work, and accumulate human capital over time. In this sense, our framework relates to recent quantitative dual-economy models with human capital accumulation, such as [Ma et al. \(2024\)](#) and [Amaral and Rivera-Padilla \(2024\)](#). Our main departure is that we incorporate a local learning externality: workers learn from peers in their sector, and they do not internalize that their location decision affects the learning opportunities of others by changing the peer skill composition within each sector and labor market. This mechanism connects our framework to classic work on

¹[Gollin \(2014\)](#) offers a recent review of the Lewis model and the dual economy concept.

knowledge spillovers and social learning (Lucas, 2004; Lucas and Moll, 2014) and to empirical and quantitative studies of learning from coworkers and local interactions (Jarosch et al., 2021; Herkenhoff et al., 2024; Crews, 2024; Lhuillier, 2024). This externality drives the different patterns of skill accumulation across sectors.

We characterize the steady-state equilibrium of our heterogeneous-worker framework and highlight two key predictions. In equilibrium, workers sort both across labor markets and within markets across sectors. Across markets, the invariant distribution exhibits positive assortative matching, as higher-skill workers concentrate in more productive labor markets due to the complementarity between worker skill and market productivity in production. Within each labor market, higher-skill workers select into the modern sector, where the scalable constant-returns technology makes high productivity particularly valuable despite regulatory costs. In contrast, lower-skill workers find it optimal to operate in the traditional sector, where decreasing returns are less binding at small scale and regulations are avoided. In addition to this sorting pattern, the model generates a dynamic modern-sector wage premium: when learning is sufficiently segmented, modern-sector workers are more exposed to higher-skill peers, accumulate human capital faster, and therefore experience steeper wage growth than traditional-sector workers, even within the same labor market.

We then examine the model’s main predictions using panel data on workers from Chile. As an important developing economy, Chile features large and segmented sectors that map naturally onto the model’s dual-economy structure. We use the Encuesta de Protección Social (EPS), a nationally representative longitudinal survey that tracks about 17,000 workers over 2002–2016. We classify workers into modern and traditional sectors based on their formality status. Informal workers lack formal labor contracts and are not subject to payroll taxes, consistent with our modeling of the traditional sector. In addition, they face informality costs that rise with output, such as expected fines or avoidance expenses (Perry et al., 2007; Ulyssea, 2020b), which imply diminishing returns to production in the traditional sector. Accordingly, we classify modern-sector workers as those affiliated with pension funds and holding a labor contract. The EPS also provides complete formal and informal job histories dating back to 1980, which allows us to follow workers across sectors over their careers. With these data in hand, we test the model’s three main predictions.

First, we show that higher-skilled workers select into the modern sector. Since skill is not directly observed in the data, we use workers’ wage outcomes as a proxy for skill, and interpret persistent wage differences across workers as reflecting differences in ability. Controlling for observable characteristics, modern-sector workers in Chile earn 17% more than their traditional-sector counterparts. However, once we account for worker fixed effects, we find that more than 40% of this wage premium is explained by unobserved worker heterogeneity. Worker fixed effects are positively correlated with modern-sector participation, indicating that modern-sector workers have higher unobserved average skill.

Second, we provide evidence for the model’s dynamic modern-sector wage premium. We analyze individual wage trajectories and examine how wage growth relates to modern-sector participation, controlling for worker demographics, industry, occupation, and firm size. We find that modern-sector workers experience larger wage increases over time than traditional-sector workers. After

one year, modern-sector workers earn 2.4% more than their traditional-sector counterparts, with the effect peaking at 8% after four years. Using wage trajectories of switchers, workers who change sectors, we show that this dynamic modern-sector premium persists even after accounting for worker fixed effects. Workers who transition from a traditional-sector to a modern-sector job experience a wage increase that is 4% higher over four years than that of workers who remain in the traditional sector. This dynamic wage premium helps explain why modern-sector workers' wages rise over the life cycle while traditional-sector workers' wages stagnate.²

Third, we find that learning from better-paid workers in the same labor market contributes to the dynamic modern-sector wage premium. Guided by our framework, we use information on workers' residence, industry, and occupation to define the labor markets in which workers interact. Then, in the spirit of [Jarosch et al. \(2021\)](#) and [Herkenhoff et al. \(2024\)](#), we assess whether exposure to higher-earning peers predicts higher future wage growth. We find that such exposure in the same labor market significantly increases future wage growth, and that this effect holds across industries, occupations, and firm-size brackets. Moreover, consistent with our sorting results, modern-sector workers tend to be more exposed to these higher-earning peers. Taken together, these findings suggest that peer learning, and the concentration of high-skilled workers in the modern sector, are key drivers of the dynamic modern-sector wage premium.

After confirming the main model predictions, we estimate the model to match relevant empirical moments in the Chilean data. We recover the parameters governing worker exit, labor-market mobility, and the initial skill distribution through minimum-distance estimators that target the empirical age distribution, the fraction of workers who switch labor markets, and the tail of the entry-wage distribution. We estimate the remaining parameters by Simulated Method of Moments. These parameters govern the traditional-sector technology, the intensity of peer learning, the segmentation of learning across sectors, and the volatility of skills, and we identify them by targeting sector employment shares together with sector and within-sector wage-quintile transition probabilities. The estimated model uncovers strong learning segmentation: modern-sector workers are three times more likely to learn from a modern-sector peer than from a traditional one, and traditional-sector workers are almost four times more likely to learn within their own sector. Combined with the sorting of high-skill workers into the modern sector, this segmentation implies that traditional-sector workers accumulate human capital substantially more slowly.

In the final section, we use the estimated model to conduct counterfactual exercises exploring the role of learning from peers on aggregate outcomes. Our first counterfactual reduces the cost of operating in the modern sector by lowering payroll taxes. Because higher-skill workers sort into the modern sector, reducing this cost raises the static payoff of modern-sector work and, at the same time, reshapes learning opportunities as the sector's skill composition changes. Eliminating payroll taxes raises the share of modern-sector workers from 59% to 80% and increases aggregate income by roughly 0.4%, reflecting both a static gain for incumbent modern-sector workers and a dynamic gain for workers who switch sectors and gain exposure to more skilled peers. The direct welfare gains from improved learning reach 1% in consumption-equivalent terms. These aggregate

²We classify modern-sector jobs as formal and traditional-sector jobs as informal. To the best of our knowledge, these patterns have not been documented previously in the informality literature.

gains, however, mask substantial heterogeneity rooted in the learning externality: workers who move into the modern sector gain more than 3%, whereas the most skilled incumbents lose about 1% as lower-skill entrants crowd into their sector and dilute their learning environment.

Our second counterfactual asks how much aggregate income and welfare would change if traditional-sector workers faced the same learning opportunities as modern-sector workers. This is not a policy experiment but a measure of the aggregate cost of the learning barriers that traditional-sector workers confront. Equalizing learning opportunities raises welfare by about 0.7%, with low-skill traditional-sector workers gaining around 1.5%. In contrast to the first exercise, modern-sector workers do not lose: because the traditional sector now offers comparable learning, some marginal workers leave the modern sector to earn higher wages, which raises the average skill, and thus the learning environment, of those who remain. This exercise underscores that the traditional sector need not be merely a stepping stone toward modern-sector work, and that it is the segmentation of learning, rather than the existence of the traditional sector itself, that constrains development.

Related Literature. Our work contributes to four strands of the literature. First, our work is most closely related to the literature on learning from peers. We generalize the ideas in [Lucas \(2004\)](#) by allowing endogenous learning environments in both the modern and traditional sectors, with interactions both within and across sectors, and we provide empirical evidence for this peer-learning mechanism. We are related to the subsequent endogenous growth frameworks of [Lucas and Moll \(2014\)](#), [Perla and Tonetti \(2014\)](#), and [Akcigit et al. \(2018\)](#), but we depart from them by emphasizing how worker sorting across sectors shapes aggregate income in developing economies. We also highlight the idea that workers learn from better peers, as in [Jarosch et al. \(2021\)](#) and [Herkenhoff et al. \(2024\)](#), with the distinction that learning in our framework is not limited to the firm. In this sense, we are closer to [Crews \(2024\)](#) and [Lhuillier \(2024\)](#), who study worker sorting across cities with local learning externalities. Closest to our empirical exercise, [Lhuillier \(2024\)](#) estimates learning from surrounding workers by projecting future wage growth on the wages of nearby individuals in French cities. We implement a similar design across labor markets and sectors in a developing economy. Finally, labor regulations associated with operating in the modern sector generate an endogenous productivity cutoff that separates workers across sectors, akin to the exporting-related sorting mechanism in [Perla et al. \(2021\)](#).

Second, we relate to papers that study how human capital accumulation shapes wage dynamics in developing economies. Building on [Lagakos et al. \(2018\)](#), [Ma et al. \(2024\)](#) show that firm-provided training in developing economies accelerates workers' skill formation, helping explain steeper age-wage profiles relative to developing economies. We differ from their framework by emphasizing learning from peers, our local externality, as the fundamental driver of differences in human capital accumulation between the modern and traditional sectors, which can also help rationalize the observed differences in wage-age profiles across developed and developing economies. Along these lines, [Engbom \(2022\)](#) finds that countries with higher rates of job-to-job transitions exhibit steeper wage profiles, which contrasts with [Donovan et al. \(2023\)](#), who document that workers in developing economies transition more across jobs without moving into better-paying ones. Abstracting from labor market frictions, we offer a reconciling explanation for these findings by showing how segmented labor markets in developing countries hinder human capital accumulation, generating

flatter wage profiles in the aggregate.

Third, our work also connects to the literature on how technological differences across sectors affect aggregate productivity in the context of economic development.³ [Lagakos \(2016\)](#) shows how retail firms in low-income countries optimally adopt traditional technologies with lower measured productivity when ownership of durable goods is low. [Duarte and Restuccia \(2019\)](#) and [Schwartzman \(2025\)](#) study structural transformation from traditional to modern services, showing that the reallocation of demand and workers across these segments is quantitatively important for aggregate income differences. We complement their results by showing that worker reallocation across traditional and modern sectors can also lead to higher aggregate income by improving the learning opportunities of lower-skill workers in the traditional sector. Closer to our work are [Herrendorf and Schoellman \(2018\)](#) and [Amaral and Rivera-Padilla \(2024\)](#), who study how complementarities between workers' human capital investments and firms' technology amplify cross-country income differences. [Engbom et al. \(2025\)](#) shows how low education levels in developing countries give rise to an endogenous dual economy with a prevalent traditional-sector organization of production. We depart from these studies by allowing for continuous skill accumulation over the life cycle, rather than a one-time human capital investment, and by emphasizing how peer composition within modern and traditional sectors shapes learning and wage dynamics.

Finally, by linking the modern and traditional sectors to formality status, we contribute to the literature on informal labor markets and their impact on aggregate outcomes. [Ulyssea \(2020b\)](#) provides a comprehensive review of recent studies analyzing firms and workers. [Ulyssea \(2018\)](#) develops a firm dynamics model with intensive and extensive margins of informality, while [Dix-Carneiro et al. \(2021\)](#) evaluates trade gains in economies with large informal labor markets. Both papers use static frameworks that abstract from worker human capital dynamics. In contrast, [Meghir et al. \(2015\)](#) considers a dynamic wage-posting setting but assumes homogeneous and time-invariant worker skills. We depart from these studies by emphasizing wage dynamics arising from ongoing skill accumulation through peer learning. More closely related to our approach are [Lopez Garcia \(2015\)](#) and [Bobba et al. \(2022\)](#), who link informality to schooling investments, and [Akcigit et al. \(2024\)](#) and [Lopez-Martin \(2019\)](#), who study informality's implications for growth. In contrast, we abstract from growth and one-time schooling decisions, and instead emphasize continuous skill accumulation and sectoral mobility over the life cycle, which shape workers' choices between the modern-formal and traditional-informal sectors.⁴

The rest of the paper is organized as follows. Section 2 outlines our framework and its main predictions. Section 3 describes the data and empirical tests of the model's predictions. Section 4 details our quantification strategy. Section 5 presents our counterfactual exercises, and Section 6 concludes.

³[Herrendorf et al. \(2014\)](#) provides a detailed overview of this literature in the broader context of structural transformation, and [Gollin and Rogerson \(2025\)](#) highlights the more recent use of heterogeneous-agent macro models in this literature, as we do.

⁴[Ulyssea \(2020a\)](#) emphasizes that we still have limited understanding of how workers make forward-looking decisions about formality status.

2 A Model of Peer Learning in a Dual Economy

In this section, we present a dynamic model of worker sorting across labor markets and between modern and traditional sectors, in which human capital accumulates through learning from peers. We then characterize the stationary equilibrium and derive predictions for worker sorting and wage growth across the two sectors. We test these predictions using Chilean panel data in Section 3.

2.1 Environment

Time is continuous and indexed by $t \geq 0$. The economy is populated by a unit mass of workers who work in one of N labor markets, indexed by $n \in \{1, \dots, N\}$. Labor markets differ in a productivity term, A_n , which determines the efficiency with which a homogeneous good is produced. Each labor market hosts two sectors, a *modern* sector (M) and a *traditional* sector (T), which differ in the technologies they operate and the regulations they face.

Workers are heterogeneous in their human capital, $z \in \mathbb{R}_+$, which determines the efficiency units of labor they supply. At any instant, a worker is described by the triple (n, S, z) , consisting of her labor market n , her sector $S \in \{M, T\}$, and her human capital z . Workers are infinitely lived but exit the economy when a Poisson death shock arrives with intensity $\delta > 0$. Each worker who exits is replaced by a newborn, who draws initial human capital z from an exogenous distribution $\psi(z)$ and is free to choose the labor market in which to begin her career. Workers discount the future at rate $\rho > 0$, derive flow utility $u(c)$ from consumption, and cannot save. Therefore, consumption equals current labor income, inclusive of a proportional government subsidy defined below.

We denote by $N_n^S(z, t)$ the mass of workers in labor market n and sector S with human capital z at time t . The mass of workers with human capital z in market n is $N_n(z, t) = N_n^M(z, t) + N_n^T(z, t)$, and the total population of the market is $L_n(t) = \int_0^\infty N_n(z, t) dz$, with sectoral counterparts $L_n^S(t) = \int_0^\infty N_n^S(z, t) dz$. These cross-sectional distributions are endogenously determined by workers' labor-market and sectoral choices.

2.1.1 Production and Wages

There are two perfectly competitive representative firms in each labor market, one in the modern sector and one in the traditional sector. Both firms produce a homogeneous good that is freely traded across markets and serves as the *numeraire*. The two firms differ in the technologies they operate. A modern-sector firm in market n combines workers in proportion to their human capital, whereas a traditional-sector firm combines workers' human capital through a concave schedule $\phi(\cdot)$:

$$Y_n^M = A_n \int_0^\infty z \ell_n^M(z) dz, \quad Y_n^T = A_n \int_0^\infty \phi(z) \ell_n^T(z) dz,$$

where $A_n > 0$ is the productivity of labor market n , and $\ell_n^S(z)$ is the mass of workers with human capital z employed in sector S . Both technologies are linear in labor, so conditional on human capital z , the two sectors differ only in how effectively they translate skill into output. The modern sector is linear in each worker's skill, whereas the traditional sector exhibits diminishing returns

to worker skill, capturing small-scale production and subsistence self-employment. Formally, we assume that $\phi(0) = 0$, $\phi(z) \geq 0$, $\phi'(z) > 0$, and $\phi''(z) < 0$ for all z , and that ϕ satisfies the Inada conditions $\lim_{z \rightarrow 0} \phi'(z) = \infty$ and $\lim_{z \rightarrow \infty} \phi'(z) = 0$.

Labor regulations differ across sectors. The modern-sector firm faces a payroll tax $\tau \geq 0$, while the traditional-sector firm avoids it. Since labor markets are perfectly competitive, firms pay each worker her marginal product of labor, net of regulatory costs, leading to the wage schedules

$$w_n^M(z) = \frac{A_n}{1 + \tau} z, \quad \text{and} \quad w_n^T(z) = A_n \phi(z), \quad (1)$$

where $w_n^M(z)$ and $w_n^T(z)$ denote the wages of workers with skill z in labor market n in the modern and traditional sectors, respectively. We assume that payroll tax revenues are rebated by a central authority to all workers, regardless of their sector, through a proportional wage subsidy at rate $b(t)$: a worker earning wage w consumes $(1 + b(t)) w$.

There are two properties of the wage schedules in (1) worth highlighting. First, within a labor market, modern- and traditional-sector wages satisfy a single-crossing property. At low skill levels, traditional-sector wages are higher than modern-sector wages because of both technological differences in production, captured by $\phi(\cdot)$, and the presence of labor regulations in the modern sector. At high skill levels, wages in the modern sector are higher than in the traditional sector because the former increase linearly with worker skill, while the latter increase at a decreasing rate due to the concavity of $\phi(\cdot)$. Hence, absent any dynamics, the wage schedules in (1) imply that, within labor markets, high-skill workers sort into the modern sector and low-skill workers into the traditional sector. Of course, workers care not only about static returns. In Section 2.5, we show how this sorting pattern extends once we account for human capital accumulation.

Second, we assume that firms are perfectly competitive in the sense that they take wages as given. Since production in each sector exhibits constant returns to scale in the number of workers of a given skill level, firms pay each worker her marginal product, leading to the wage schedules in (1). This notion of perfect competition contrasts with Jarosch et al. (2021), who assume that firms can price the learning opportunities they provide to workers. In their framework, wages reflect not only workers' marginal products but also the value of learning at the firm. In contrast, as we explain in Section 2.1.3, learning in our framework occurs at the labor-market level. Hence, we assume that firms cannot price workers' learning opportunities. This assumption implies that wages reflect only technological differences across sectors and labor markets, rather than differences in the learning environments they provide.

2.1.2 Labor Market Mobility

Workers move across labor markets subject to mobility frictions and costs. First, they cannot move instantly across markets. Instead, they remain in their current labor market until they receive a moving opportunity, which arrives according to a Poisson process with intensity λ . Second, when

an opportunity arrives, a worker observes a vector of idiosyncratic location tastes $\{\varepsilon_m\}_{m=1}^N$, which we assume are independently and identically distributed across workers and markets according to a Type-I extreme value distribution with scale $1/\nu$. The parameter ν governs the dispersion of tastes: as $\nu \rightarrow \infty$, tastes vanish and all workers locate in the market offering the highest continuation value, while a smaller ν spreads workers more evenly across markets. After observing the collection of location tastes, a worker in market n may move to any market $m \in \{1, \dots, N\}$ by paying a bilateral mobility cost ξ_{nm} , with $\xi_{nn} = 0$.

We assume that workers can freely switch sectors within a labor market. Accordingly, we denote by $V_n(z, t)$ the value of a worker with skill z in market n at time t . We later formally define $V_n(z, t)$ as the upper envelope of the sector-specific values.

Conditional on receiving a relocation opportunity, a worker chooses the market that maximizes her continuation value net of mobility costs and idiosyncratic tastes. The continuation value of labor market mobility is

$$\mathcal{M}_n[V](z, t) = \lambda \left(\mathbb{E}_\varepsilon \left[\max_m \{V_m(z, t) - \xi_{nm} + \varepsilon_m\} \right] - V_n(z, t) \right),$$

where the expectation is taken over the idiosyncratic location tastes, ε_m . Under the extreme value assumption, this continuation value takes the closed form

$$\mathcal{M}_n[V](z, t) = \lambda \left[\frac{1}{\nu} \log \left(\sum_{m=1}^N e^{\nu(V_m(z, t) - \xi_{nm})} \right) - V_n(z, t) \right], \quad (2)$$

and the probability that a worker with skill z who receives an opportunity in market n relocates to market m is given by

$$m_{nm}(z, t) = \frac{e^{\nu(V_m(z, t) - \xi_{nm})}}{\sum_{k=1}^N e^{\nu(V_k(z, t) - \xi_{nk})}}.$$

These shares, together with the death-and-birth process, govern the reallocation of workers across markets in the distributional dynamics below.

2.1.3 Human Capital Accumulation

Workers accumulate skill by learning from other workers in their labor market. However, learning is segmented across sectors and depends on the skill distribution of workers in each sector. Formally, human capital for a worker in market n and sector S follows a geometric Brownian motion:

$$d \log z_n^S(t) = \gamma Z_n^S(t) dt + \sigma_S dW(t), \quad (3)$$

where $\gamma > 0$ governs the importance of learning from peers, $Z_n^S(t)$ is a measure of the stock of

human capital in sector S and market n , which we call *vibrancy* and define below, $W(t)$ is a standard Wiener process, and $\sigma_S > 0$ is a sector-specific volatility. The skill evolution equation (3) has an endogenous drift component, which depends on $Z_n^S(t)$, and an exogenous volatility component, disciplined by σ_S , which captures sector-specific shocks to skills.⁵

The *vibrancy* of market n and sector S is defined as

$$Z_n^S(t) = \left[\int_0^\infty z^\theta g_n^S(z, t) dz \right]^{\frac{1}{\theta}}, \quad (4)$$

where $\theta \geq 1$ and $g_n^S(z, t)$ is the effective peer-skill distribution faced by workers in market n and sector S , which we specify below.⁶ The parameter θ governs how different parts of the skill distribution combine in fostering human capital accumulation. In general, vibrancy is higher in environments populated by more skilled peers. When $\theta = 1$, vibrancy is the mean of peer human capital, so skills are perfect substitutes in learning. As $\theta \rightarrow \infty$, vibrancy is determined by the human capital of the most skilled worker. Thus, θ controls the extent to which learning depends on the upper tail of the peer-skill distribution within sectors and labor markets.

The effective peer-skill distribution $g_n^S(z, t)$ is a mixture distribution that places weight π^S on workers in sector S and weight $1 - \pi^S$ on workers in the other sector, which we denote by $-S$. Hence, the effective peer-skill distribution takes the form

$$g_n^S(z, t) = \frac{\pi^S N_n^S(z, t) + (1 - \pi^S) N_n^{-S}(z, t)}{\pi^S L_n^S(t) + (1 - \pi^S) L_n^{-S}(t)}. \quad (5)$$

This expression is a well-defined density over workers' skill in market n and sector S : $g_n^S(z, t) \geq 0$ for all z , and $\int_0^\infty g_n^S(z, t) dz = 1$.

The intuition behind (5) is as follows. Workers in market n can potentially be exposed to any worker with skill level z in the same labor market. However, exposure is not uniform across sectors. The parameter π^S captures the degree of *learning segmentation* by measuring how biased the interactions of workers in sector S are toward their own sector. When $\pi^S = 1$, workers in sector S learn only from workers in their own sector, so learning is fully segmented. When $\pi^S = 0$, workers in sector S learn only from workers in the other sector. Finally, when $\pi^S = 1/2$, workers are equally exposed to both sectors and learning is fully integrated.

2.2 Worker Values and Labor Supply

We now characterize workers' dynamic labor supply decisions. A worker in market n and sector S receives flow utility $u((1 + b(t)) w_n^S(z))$ from consuming her subsidized wage. Over time, she

⁵One example of such sector-specific skill shocks could be differences in working conditions across sectors.

⁶Our general theory also admits $0 < \theta < 1$. However, restricting attention to $\theta \geq 1$ simplifies the sufficient conditions for a sorting equilibrium below.

accumulates human capital according to (3), can reallocate across labor markets when a moving opportunity arrives, and exits the economy when a death shock arrives.

To write the worker's problem, it is useful to express the law of motion for human capital in levels. By Itô's lemma, z evolves as

$$dz_n^S(t) = \mu_n^S(t) z dt + \sigma_S z dW(t), \quad \mu_n^S(t) \equiv \gamma Z_n^S(t) + \frac{1}{2} \sigma_S^2. \quad (6)$$

With this notation, the value of a worker with skill z in market n satisfies the Hamilton-Jacobi-Bellman (HJB) variational inequality

$$(\rho + \delta) V_n(z, t) - \partial_t V_n(z, t) = \max_{S \in \{M, T\}} \left\{ u((1 + b(t)) w_n^S(z)) + \mu_n^S(t) z \partial_z V_n(z, t) + \frac{\sigma_S^2}{2} z^2 \partial_{zz} V_n(z, t) \right\} + \mathcal{M}_n[V](z, t). \quad (7)$$

The left-hand side of (7) is the effective discounting of the worker's value, adjusted for time variation in the value function. The first term on the right-hand side is the flow value of consumption. The second term captures the continuation value from the expected growth of human capital, and the third term captures the effect of skill volatility on the worker's value. The last term is the option value of mobility across labor markets.

How do workers choose which sector to work in? Since there are no within-market reallocation costs, sector choice is embedded directly in the worker's dynamic problem and the labor supply decision $S_n^*(z, t)$ attains the maximum in (7).

The optimal sectoral choice, together with the choice of destination market embedded in the mobility shares $m_{nm}(z, t)$, fully summarizes workers' policies. We characterize this labor supply decision in Section 2.5.

2.3 Skill Distribution Dynamics

We now derive the law of motion for the cross-sectional distribution of workers' human capital across markets and sectors. First, the flux of mass through skill level z in n and S is

$$J_n^S(z, t) = (\mu_n^S(t) - \sigma_S^2) z N_n^S(z, t) - \frac{\sigma_S^2}{2} z^2 \partial_z N_n^S(z, t).$$

Let $\mathbb{1}_n^S(z, t)$ be an indicator equal to one if sector S is optimal for a worker with skill z in market n , as implied by (7), and zero otherwise. Then, the mass of workers with skill level z in labor market n and sector S evolves according to the Kolmogorov Forward Equation (KFE)

$$\begin{aligned} \partial_t N_n^S(z, t) = & -\partial_z J_n^S(z, t) + \lambda \left(\sum_{k=1}^N m_{kn}(z, t) N_k(z, t) \mathbb{1}_n^S(z, t) - N_n^S(z, t) \right) \\ & + \delta [\psi(z) \Pi_n(z, t) \mathbb{1}_n^S(z, t) - N_n^S(z, t)]. \end{aligned} \quad (8)$$

The KFE (8) combines three forces. The first term, $-\partial_z J_n^S$, captures the endogenous drift and diffusion of human capital through learning. The second term is the net reallocation due to labor market mobility. Finally, the third term captures death turnover. Workers die at rate δ , while newborns arrive with human capital drawn from $\psi(z)$ and start in market n with probability

$$\Pi_n(z, t) = \frac{e^{\nu V_n(z, t)}}{\sum_{k=1}^N e^{\nu V_k(z, t)}}. \quad (9)$$

After choosing a market, newborns enter the sector that maximizes their value in that market. Equation (9) reflects that newborns, like incumbent workers who relocate, draw idiosyncratic location tastes and sort into the market that offers them the highest value.

The system is closed by a boundary condition stating that mass vanishes in the upper tail,

$$\lim_{z \rightarrow \infty} N_n^S(z, t) = 0, \quad S \in \{M, T\}.$$

2.4 Equilibrium Definition

We are interested in a stationary equilibrium of this economy in which the cross-sectional distributions are time invariant, $\partial_t N_n^S(z, t) = 0$, and so are the value functions, $\partial_t V_n(z, t) = 0$. We therefore drop the time index from all equilibrium objects. Although individual workers relocate across markets, switch sectors, and accumulate skill over their lives, aggregate objects and the joint distribution of workers across markets, sectors, and human capital remain unchanged.

A stationary equilibrium consists of wages $w_n^M(z)$ and $w_n^T(z)$, value $V_n(z)$, labor supply decisions $S_n^*(z)$, sectoral masses $N_n^M(z)$ and $N_n^T(z)$, effective peer-skill distributions $g_n^M(z)$ and $g_n^T(z)$, vibrancies Z_n^M and Z_n^T , and subsidy rate b , such that:

- (i) *Optimal labor demand.* Firms maximize profits, so wages satisfy (1).
- (ii) *Workers' dynamic problem.* The value $V_n(z, t)$ and labor supply $S_n^*(z)$ are given by (7).
- (iii) *Learning consistency.* The effective peer-skill distributions $g_n^S(z)$ and the vibrancies Z_n^S are given by (4) and (5).
- (iv) *Distribution dynamics.* The sectoral masses evolve according to the Kolmogorov forward equation (8), together with the boundary condition that mass vanishes in the upper tail.
- (v) *Government balanced budget.* The government runs a balanced-budget period by period,

$$b \sum_{n=1}^N \sum_{S \in \{M, T\}} \int_0^\infty w_n^S(z) N_n^S(z) dz = \sum_{n=1}^N \int_0^\infty \tau w_n^M(z) N_n^M(z) dz.$$

- (vi) *Labor market clearing.* Labor demand equals labor supply in every market and sector, $\ell_n^S(z) =$

$N_n^S(z)$, and the population adds up to the unit mass of workers,

$$\sum_{n=1}^N L_n(t) = 1.$$

The definition of equilibrium highlights the local learning externality. Individual workers take vibrancies as given when making their labor supply decisions. Nevertheless, these vibrancies are endogenously determined by the resulting mass of workers across markets and sectors. In turn, workers' optimal sectoral choices affect the human capital evolution of others.

We solve for the stationary equilibrium numerically using a finite-difference scheme, following [Achdou et al. \(2022\)](#). Appendix B provides a detailed description of the algorithm.

2.5 Equilibrium Characterization

We now characterize the stationary equilibrium in three steps. First, conditional on the existence of values and an invariant distribution, we provide conditions under which workers sort across sectors, with more productive workers selecting into the modern sector. Second, we show that, under additional conditions, such a *sorting equilibrium* exists. Third, we discuss uniqueness. All proofs, together with auxiliary lemmas, are relegated to Appendix A.

From (7), it is useful to define the *flow advantage* of the modern sector,

$$\Delta_n(z) \equiv \underbrace{u((1+b)w_n^M(z)) - u((1+b)w_n^T(z))}_{\text{static gain}} + \underbrace{(\mu_n^M - \mu_n^T) z V_n'(z) + \frac{\sigma_M^2 - \sigma_T^2}{2} z^2 V_n''(z)}_{\text{dynamic gain}}, \quad (10)$$

so that $S_n^*(z) = M$ exactly when $\Delta_n(z) \geq 0$. The static gain compares the flow utility workers obtain across sectors, which is determined by the potential wages in each sector. The dynamic gain captures differences in the human capital dynamics offered by each sector. First, sectors differ in expected skill growth, pinned down by μ_n^M and μ_n^T . These expected gains are valued through the slope of the value function. Second, sectors expose workers to different skill shocks, determined by σ_M^2 and σ_T^2 , which are valued through the curvature of the value function. Worker sorting takes a cutoff form in market n when $\Delta_n(z)$ crosses zero only once, from below.

We first introduce two functional forms assumptions for utility and returns in the traditional sector under which we obtain sharp sorting results. Importantly, these are the same functional form assumptions that we impose in the quantitative analysis in Section 4.

Assumption 1 (Functional forms). Workers have log utility, $u(c) = \log c$, and the traditional technology is $\phi(z) = z^\eta$, with $\eta \in (0, 1)$.

Under Assumption 1, we can characterize the sorting cutoff in the static version of our model. Let

$$\hat{z} \equiv (1 + \tau)^{\frac{1}{1-\eta}}. \quad (11)$$

Then, from (10), absent any dynamic gains, workers with $z \geq \hat{z}$ select into the modern sector, while workers with $z < \hat{z}$ select into the traditional sector. Intuitively, this static cutoff is increasing in τ : higher payroll taxes in the modern sector depress wages in that sector, making it less attractive for marginal workers. Similarly, the cutoff is increasing in η : a more scalable traditional-sector technology increases wages in the traditional sector, making it more appealing for marginal workers. The static cutoff is the same across all markets.

How can the static sorting equilibrium be extended to the dynamic setting? The expression in (10) suggests that, as long as the modern sector provides better learning opportunities than the traditional sector in each market, $Z_n^M \geq Z_n^T$, and the volatility of skill shocks does not differ too much across sectors, the static sorting result should extend to the dynamic learning environment.⁷ Assumption 2 introduces a regularity condition on the learning segmentation parameters that helps ensure the former condition holds.

Assumption 2 (Consistent segmentation). The segmentation parameters satisfy $\pi^M + \pi^T \geq 1$.

Assumption 2 rules out learning interactions that are biased toward the opposite sector on net. It holds, in particular, whenever workers in each sector are at least as likely to interact with peers in their own sector as with peers in the other sector, $\pi^M \geq 1/2$ and $\pi^T \geq 1/2$, including the case of random interactions, $\pi^M = \pi^T = 1/2$.⁸

Finally, to ensure that vibrancies across markets are finite and human capital does not grow explosively, we impose a regularity condition on the skill distribution of new workers.

Assumption 3 (Turnover dominates learning). Let $m_\psi(\theta)$ be the θ -th moment of the entry distribution $\psi(\cdot)$ and let $\sigma \equiv \max\{\sigma_M, \sigma_T\}$. Then, $\theta^2 \sigma^2 / 2 < \delta$ and

$$m_\psi(\theta) \leq \kappa \frac{\left(\delta - \theta^2 \frac{\sigma^2}{2}\right)^{\theta+1}}{(\theta+1)^{\theta+1} \gamma^\theta \delta}, \quad (12)$$

where $\kappa \in (0, 1]$ is a constant defined in Appendix A.

Assumption 3 requires turnover to be sufficiently high relative to the learning process and the upper tail of the entry skill distribution. In our quantitative exercise, the entry distribution is Pareto with unit scale and shape parameter ϑ , so that $m_\psi(\theta) = \vartheta/(\vartheta - \theta)$ provided $\vartheta > \theta$. Assumption 3 then reduces to a lower bound on the Pareto shape parameter, $\vartheta \geq \theta R/(R - 1)$, where R denotes the right-hand side of (12). Hence, the newborn skill distribution must be sufficiently thin-tailed for a stationary equilibrium to exist. Importantly, this condition is more likely to hold when turnover δ is high or the importance of learning from peers, γ , is sufficiently low.

Assumption 3 also has a natural mathematical interpretation. It is a drift condition in the sense of Meyn and Tweedie (1993): worker turnover pulls the skill distribution back toward the entry distribution faster than peer learning spreads it out.

⁷This intuition anticipates that the solution to (7) is strictly increasing and concave in z .

⁸Assumption 2 turns out to be the empirically relevant configuration, since our estimates in Section 4, which do not impose this assumption, reveal a strong own-sector bias in both sectors.

Proposition 1 (Existence of a Sorting Equilibrium). Suppose Assumptions 1–3 hold. Then, there exists $\Sigma > 0$ such that, whenever $|\sigma_M - \sigma_T| < \Sigma$, a stationary equilibrium with positive sorting exists: there are finite cutoffs $\bar{z}_n$ such that $S_n^*(z) = T$ for $z < \bar{z}_n$ and $S_n^*(z) = M$ for $z \geq \bar{z}_n$. Moreover, if learning is sufficiently segmented, $\pi^M > 1/2$ and $\pi^T > 1/2$, then $\bar{z}_n < \hat{z}$ for all n .

Proof. See Appendix A.2. □

Proposition 1 formalizes the main mechanism of the model. Since workers sort across sectors based on their human capital, the modern sector attracts the most skilled workers. This concentration of skilled workers makes the modern sector a better learning environment, so forward-looking workers with sufficiently high skill find it optimal to choose that sector. Sorting reinforces itself: skilled workers select into the modern sector, their presence improves learning in that sector, and better learning opportunities foster faster human capital accumulation relative to the traditional sector. Workers with very low levels of human capital, instead, value more the static gains offered by the traditional sector, at the expense of future wage growth.

This logic also explains why the dynamic cutoff lies below the static cutoff when learning is sufficiently segmented. In the static model, workers choose sectors only based on current wages. In the dynamic model, they also value future skill growth. Since the modern sector provides better learning opportunities, some workers enter the modern sector even when their current wage would be higher in the traditional sector. These marginal workers are willing to exchange lower wages today for better learning opportunities that imply higher wages in the future. Therefore, more workers select into the modern sector than static wage comparisons alone would imply. The condition that sectoral skill volatility is not too different ensures that this learning advantage is not overturned by differences in risk.

Having established the existence of a sorting equilibrium, a natural question arises: when is this equilibrium unique? Conditional on the vector of vibrancies across markets and sectors, $Z \equiv \{Z_n^M, Z_n^T\}_{n=1}^N$, all remaining equilibrium objects are unique. Values are uniquely determined by (7), the cutoff policy is unique because the flow advantage is strictly increasing in $\log z$, the invariant distribution is pinned down by (8), and the subsidy rate is pinned down by the government budget constraint. Any multiplicity must therefore arise in the finite-dimensional fixed-point problem $Z = T(Z)$. Our next result shows that when the learning externality is not too strong, this fixed point is unique as well.

Proposition 2 (Uniqueness). Suppose Assumptions 1–3 hold, and let $|\sigma_M - \sigma_T| < \Sigma$ as in Proposition 1. Then, there exists $\bar{\gamma} > 0$ such that, for all $\gamma \in [0, \bar{\gamma})$, the sorting equilibrium of Proposition 1 is the unique stationary equilibrium.

Proof. See Appendix A.2. □

Proposition 2 says that when the peer-learning channel is not too strong in determining human capital accumulation, the sorting equilibrium is unique. As is standard in models with externalities, we cannot rule out multiple equilibria when learning from peers is sufficiently strong. What can

differ across equilibria is how much human capital the sorting equilibrium can sustain. In one possible equilibrium, workers anticipate strong learning in the modern sector, enter the sector earlier in their careers, and spend more time accumulating skills at a fast rate. As a result, the stock of human capital in the modern sector remains high, validating workers' initial anticipation. In another possible equilibrium, workers anticipate weak learning, enter the modern sector only when current wages justify it, and accumulate skills more slowly. The same market then settles at a permanently lower human capital stock. Across markets, expectations and history may similarly determine which location becomes the high-vibrancy hub when fundamentals are similar.

In our framework, three forces work against this multiplicity. First, idiosyncratic tastes disperse workers across markets. Second, workers continuously exit the labor market, bringing in new workers from the common entry distribution. Third, productivity differences break symmetry across markets. In our quantitative implementation, we test for multiplicity by initializing our algorithm from different starting distributions. Reassuringly, the predictions we derive next rely only on the sorting structure in Proposition 1, and therefore hold in any sorting equilibrium.

2.6 Model Predictions

The stationary equilibrium delivers two main predictions that organize our empirical analysis. The first is the sorting equilibrium established in Proposition 1. The second is a dynamic modern-sector wage premium. Conditional on the labor market, modern-sector workers experience faster expected wage growth than traditional-sector workers, as formalized in the following lemma.

Lemma 1 (Dynamic modern-sector premium). *Consider a sorting equilibrium as in Proposition 1, and fix a market n and a horizon $h > 0$. If learning is sufficiently segmented, $\pi^M > 1/2$ and $\pi^T > 1/2$, then the expected difference in log-wage growth between modern- and traditional-sector workers in market n between t and $t + h$ is strictly positive:*

$$\mathbb{E}_t[\Delta \log w_{n,t+h}^M] - \mathbb{E}_t[\Delta \log w_{n,t+h}^T] = \gamma(Z_n^M - \eta Z_n^T) h > 0.$$

Proof. See Appendix A.2. □

Lemma 1 is the main novel testable prediction of our theory. When learning is segmented, workers in both sectors interact more frequently with peers from their own sector. Hence, vibrancies in the stationary sorting equilibrium satisfy $Z_n^M > Z_n^T$. This result reflects the fact that the modern sector attracts more skilled workers, so the modern peer-skill distribution first-order stochastically dominates the traditional one. Consequently, modern-sector workers accumulate human capital faster. Importantly, because the traditional technology is concave, a given increase in human capital raises traditional-sector wages with elasticity $\eta < 1$. Hence, although technological differences across sectors are not necessary for the dynamic premium to arise, they amplify it.

In the next section, we test Proposition 1 and Lemma 1 in the Chilean data. We first test the sorting prediction using the distribution of estimated worker skill across sectors. We then test the dynamic prediction by estimating, at each horizon h , the average difference in wage growth between modern- and traditional-sector workers.

3 Empirical Analysis

In this section, we test the main model predictions. We begin by describing the data and our definitions of modern and traditional jobs. Our analysis focuses on Chile, where we can exploit rich worker survey data from an important developing economy in which a large fraction of workers still operate in the traditional sector.

3.1 Data

Our main data source is the Chilean *Encuesta de Protección Social* (EPS). The EPS is a comprehensive longitudinal survey designed to gather data on various aspects of social protection. The survey spans five waves conducted in 2002, 2004, 2006, 2009, and 2015.⁹ We use two blocks of the survey. First, we use the block containing information on individuals' demographics, such as age and education. Second, we use the block containing individuals' job histories dating back to 1980.¹⁰ The job history includes information on job type, occupation, economic sector, region, weekly hours worked, starting and ending dates, type of contract, pension fund contributions, job category, and the number of workers in the firm. Given our definition of modern and traditional jobs below, we restrict our sample to start in the 2004 wave, the year in which individuals with and without pension affiliation were included, making the survey representative of all workers.

We define modern and traditional-sector jobs in the data in a way that is consistent with our framework. In the model, modern-sector workers are those employed by firms that comply with labor regulations. Accordingly, we define a modern-sector job as one with a formal labor contract and pension contributions, which maps to the payroll tax in the model. We define a traditional-sector job as any job that does not comply with these labor regulations. In doing so, our definitions of modern and traditional employment coincide with the standard distinction between formal and informal work used in the literature (Ulyssea, 2020b) and typically used by statistical agencies.¹¹

The EPS offers four advantages for analyzing employment dynamics among modern and traditional workers. First, the detailed information available for each job allows us to classify workers as modern or traditional based on their contract type and pension fund contributions. Second, the data on disposable monthly income and hours worked allow us to construct hourly wages for jobs starting in 2002.¹² Third, although the survey is conducted only in selected years, the job-history block allows us to reconstruct each worker's employment history since 1980. We aggregate workers' jobs to construct our baseline sample, an unbalanced annual panel with more than 17,000 workers observed between 2002 and 2016.¹³ The length and coverage of the panel allow us to track modern and traditional workers over long periods. In particular, we observe 50% of workers in our sample

⁹We exclude the 2012 and 2019 waves from our analysis. The 2012 wave is not representative at the national level, and the questions in 2019 changed because of the COVID-19 pandemic.

¹⁰When an individual enters the survey, they are asked for their job history since 1980. If, in a specific wave, the individual was already part of the survey, they are asked about their job history since the survey's last wave.

¹¹Chile's statutory labor laws state that at the start of any labor relation, any non-affiliated worker generates automatic affiliation to the pension system and the obligation to contribute 10% of their salary to the fund.

¹²We construct wages for 2002 and 2003 retroactively using information from workers' job histories.

¹³For workers with multiple jobs in a given year, we retain the job with the highest total hours worked.

Table 1: Summary Statistics

	(1) Modern	(2) Traditional	(3) All
Fraction of workers	0.70	0.30	...
Mean log wage	0.81	0.44	0.70
Std. deviation log wage	0.63	0.69	0.67
Mean weekly working hours	46.00	42.19	44.85
Fraction of male workers	0.63	0.59	0.62
Mean experience (years)	15.24	15.38	15.28
Transitions (by initial status)			
Frac. of modern workers next year	0.97	0.08	...
Frac. of traditional workers next year	0.03	0.92	...
Number of observations	86,646	35,114	121,760
Number of workers	...	...	17,260

Notes: Table 1 displays summary statistics for the baseline sample 2002-2016. We include male workers aged 15 to 65 and female workers aged 15 to 60. We exclude workers without remuneration, public-sector and armed forces employees. Wage is the real hourly wage in 2004 US dollars. The variable experience corresponds to the equivalent in years of a full-time work schedule of 48 hours per week.

for at least five years and 20% for at least ten years.¹⁴ Finally, although we restrict the analysis to jobs active from 2002 onward because of wage availability, we use workers' job histories since 1980 to measure experience in the modern and traditional sectors separately.¹⁵

We exclude certain types of workers from our analysis. First, in accordance with Chilean labor regulations, we include male workers aged 15 to 65 and female workers aged 15 to 60, aligning with the country's legal working and retirement ages for men and women, respectively. Second, in our baseline specifications, we omit unpaid workers, as well as public-sector and armed forces employees. Importantly, our data allow us to include self-employed workers and entrepreneurs, who represent a large fraction of traditional-sector workers.¹⁶

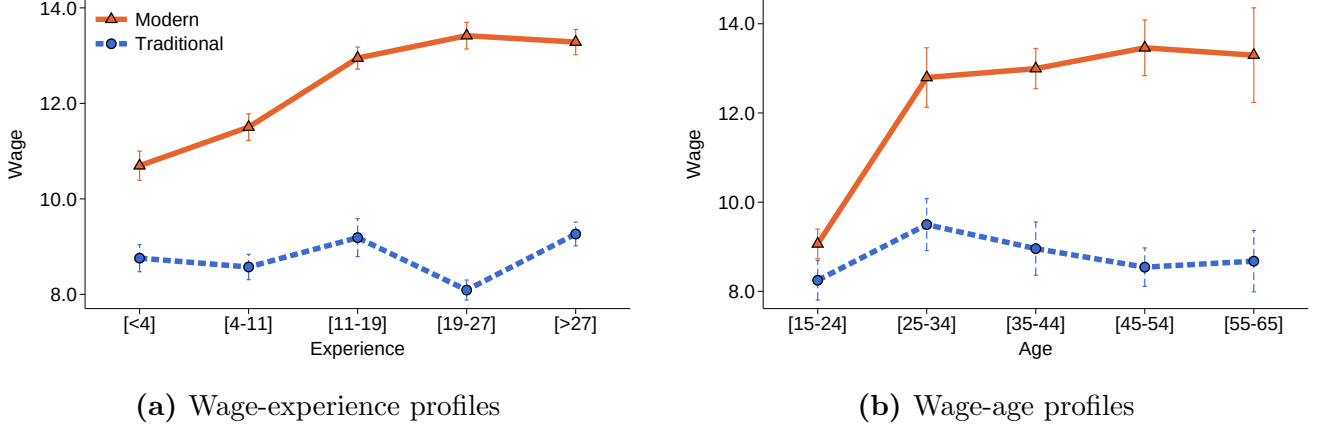
Table 1 displays summary statistics for our baseline sample. Three key points are worth highlighting. First, wages are, on average, higher for modern workers than for traditional workers. Nevertheless, wages among traditional workers are more dispersed than those among modern workers, consistent with previous findings. Second, modern and traditional workers work similar hours per week and have similar years of experience. This similarity supports our modeling choice of abstracting from labor supply decisions and from an explicit allocation of time between working

¹⁴In contrast, Meghir et al. (2015) use a rotating panel in Brazil that follows individuals for five consecutive months and then for another four months one year after their entry into the sample. Samaniego De La Parra and Fernández Bujanda (2020) use a rotating panel in Mexico that follows workers for only five quarters.

¹⁵We compute experience by dividing cumulative hours worked up to a particular year by the number of hours corresponding to a full-time schedule of 48 hours per week.

¹⁶We provide robustness exercises in which we restrict the sample to salaried workers only.

Figure 1: Wages Over the Life Cycle for Modern and Traditional Workers



Notes: Figure 1 displays wage paths over the life cycle for modern and traditional workers. Panel 1(a) shows the average wage for each experience quintile and panel 1(b) for each 10-year age bin. Vertical dashed lines denote 95 percent confidence intervals.

and learning. Third, transition probabilities between the modern and traditional sectors are highly asymmetric. On average, 9% of traditional workers move to the modern sector in the following year, whereas only 3% of modern workers move to the traditional sector. Importantly, these transition probabilities are computed by restricting the sample to active workers, thereby excluding transitions into and out of unemployment. Appendix C provides additional descriptive statistics, including the share of modern workers by age, education, occupation, industry, and firm size.

3.2 Testing the Model's Predictions

This section presents empirical evidence on wages for modern and traditional workers that is consistent with our model. We begin by documenting wage-experience profiles, which reveal different wage growth patterns across sectors. We then test the model's main predictions in two steps. First, we present evidence that high-skill workers sort into the modern sector. Second, we test for the presence of a dynamic modern-sector wage premium.

Figure 1 presents wage-experience and wage-age profiles for modern and traditional workers. The left panel displays the wage-experience profiles. We first classify workers as modern or traditional according to our definition, divide them into five experience bins, and compute the average wage within each bin and sector. The right panel displays wage-age profiles, which are constructed analogously using workers' age. Both panels reveal stark differences in life-cycle wage growth between modern and traditional workers.

Figure 1 shows substantial differences in wage levels and wage growth between modern and traditional workers over the life cycle. Focusing first on the wage-experience profiles, modern workers begin their careers with wages around 20% higher than those of traditional workers. This wage gap then widens over the life cycle, reaching more than 40% by the end of workers' careers. The divergence is mainly driven by wage growth among modern workers, while traditional workers' wages remain mostly flat over time. The wage-age profiles display a similar pattern. Although the

wage gap early in the life cycle is smaller, at around 12%, it reaches a similar magnitude by the end of workers’ careers. Interestingly, the timing of the divergence differs across the two profiles. In the wage-experience profiles, the gap widens gradually over the life cycle, whereas in the wage-age profiles most of the divergence occurs early.

These life-cycle wage patterns for modern and traditional workers align with recent findings on wage growth across countries. [Lagakos et al. \(2018\)](#) document that wages in developed countries increase more over the life cycle than wages in developing economies. Our findings suggest that the modern sector behaves like a “developed-economy labor market”, whereas the traditional sector displays patterns typical of a “developing-economy labor market.” In [Appendix C](#), we report the same wage profiles restricting the sample to salaried workers. The patterns remain consistent under this alternative specification.

We interpret these initial findings as motivating evidence for the main mechanism of our model. The initial wage differences are consistent with more productive workers sorting into the modern sector. The subsequent widening of the wage gap over the life cycle is also consistent with modern-sector workers experiencing faster human capital accumulation because of better learning opportunities. In the following subsections, we provide more direct tests of both mechanisms.

3.2.1 Worker Sorting

We investigate worker sorting across the modern and traditional sectors in two ways. First, we examine whether workers’ observable characteristics that are potentially related to skill are correlated with their sectoral choices. Perhaps not surprisingly, [Figure C2](#) in the Appendix shows that workers with higher educational attainment are more likely to be employed in the modern sector. The share of modern workers among those with no schooling is slightly above 50%. In contrast, more than 80% of workers with two or more years of college education are employed in the modern sector. To the extent that educational attainment is positively correlated with skill, these patterns support the idea that more skilled workers sort into the modern sector.

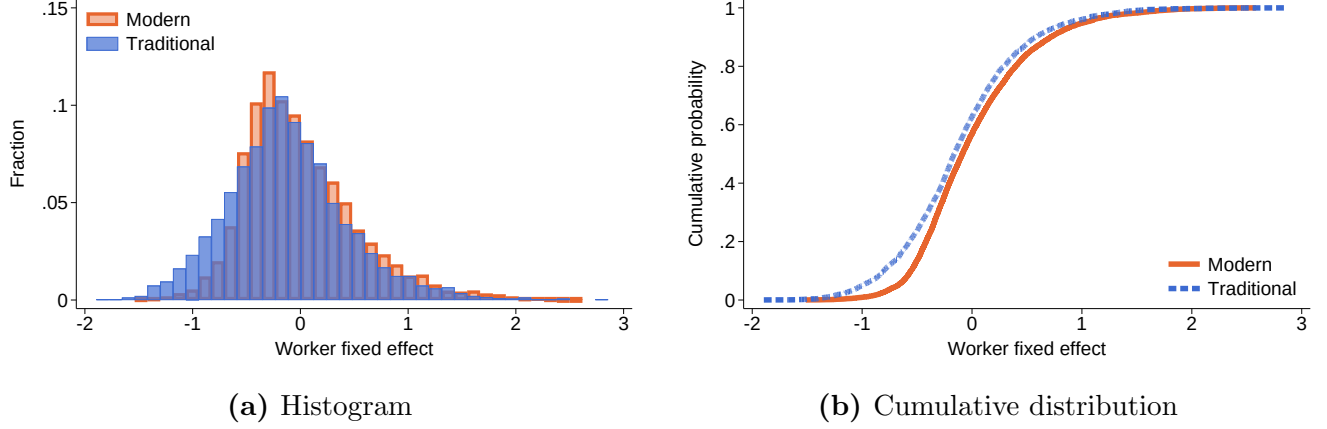
Second, we study the extent to which workers with higher unobserved skill are more likely to be employed in the modern sector. To do so, we study the modern-traditional wage gap in levels and assess the role of worker sorting in explaining this gap. We estimate the following regression:

$$\log w_{i,t} = \beta \text{Modern}_{i,t} + \Gamma X_{i,t} + \alpha_t + \alpha_i + \zeta_{i,t}, \quad (13)$$

where $\log w_{i,t}$ is the log wage of worker i at time t , $\text{Modern}_{i,t}$ is an indicator equal to one if worker i is employed in the modern sector in year t and zero if she is employed in the traditional sector, and $X_{i,t}$ is a vector of worker characteristics that includes age, education, gender, occupation, industry, region, firm size, and experience in the modern and traditional sectors. Moreover, α_t and α_i denote time and worker fixed effects, respectively.

The worker fixed effects, α_i , capture time-invariant worker skill that is rewarded in both sectors. Importantly, these fixed effects capture skill differences beyond the observable characteristics included in $X_{i,t}$, such as age, education and experience in each sector. Our framework predicts that

Figure 2: Worker Fixed Effects for Modern and Traditional Workers



Notes: Figure 2 displays the distribution of the worker fixed effects α_i estimated from equation (13), controlling for age, education, occupation, industry, region, firm size, experience in each sector, and time fixed effects. We keep one observation per worker and assign workers to the sector in which we observe them most often. Panel 2(a) shows the histogram and panel 2(b) the empirical cumulative distribution function of the worker fixed effects for modern and traditional workers.

more skilled workers sort into the modern sector. Therefore, through the lens of the model, the distribution of worker fixed effects among modern workers should lie to the right of the corresponding distribution among traditional workers.

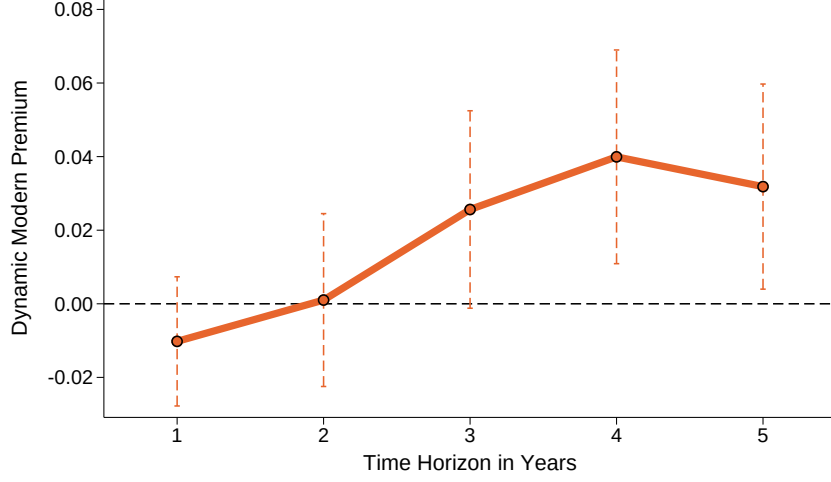
Figure 2 presents the distribution of estimated worker fixed effects for modern and traditional workers. We retain one observation per worker and assign each worker to the sector in which she is observed most frequently. The figure provides evidence of worker sorting. Panel 2(a) shows that traditional workers are overrepresented at the bottom of the skill distribution, whereas modern workers are concentrated at the top. Panel 2(b) shows that the cumulative distribution of worker fixed effects for modern workers lies everywhere below that of traditional workers. This implies that the modern-sector skill distribution first-order stochastically dominates the traditional-sector distribution. Moreover, the estimated correlation between workers' fixed effects, α_i , and the modern-sector indicator, $\text{Modern}_{i,t}$, is positive, at approximately 0.10, and statistically significant. Hence, consistent with our model, more skilled workers, sort into the modern sector.

Table D1 in Appendix D reports the estimates of β in equation (13). Modern workers earn a wage premium even after conditioning on worker fixed effects and observable characteristics. Through the lens of our model, a wage premium among workers with the same skill level provides evidence of diminishing returns to worker skill in the traditional sector.

3.2.2 Dynamic Modern-Sector Premium

We now turn to the main novel prediction of our model: the dynamic modern-sector wage premium. We test this prediction in two steps. First, we examine the short- and medium-run effects of working in the modern sector on future wages. Second, we provide evidence that learning from better-paid peers in the modern sector is a plausible mechanism generating these persistent wage differences between modern- and traditional-sector workers.

Figure 3: Modern Sector Dynamic Wage Premium



Notes: The figure displays the modern future wage premium based on the estimates of β_h from equation (14) including worker fixed effects. Each dot represents the estimated wage premium at horizon $t+h$: $\exp(\hat{\beta}_h) - 1$, for $h = 1, \dots, 5$. Controls include age, education, gender, occupation, industry, region, experience, wage decile, and time and worker fixed effects. Vertical dotted lines denote 95 percent confidence intervals.

We begin by studying how employment in the modern sector relates to future wage growth. To do so, we estimate a variant of (13) that relates workers' future wage growth to their sectoral status and a set of observable and unobservable characteristics. Specifically, we estimate the following relationship separately for different time horizons h :

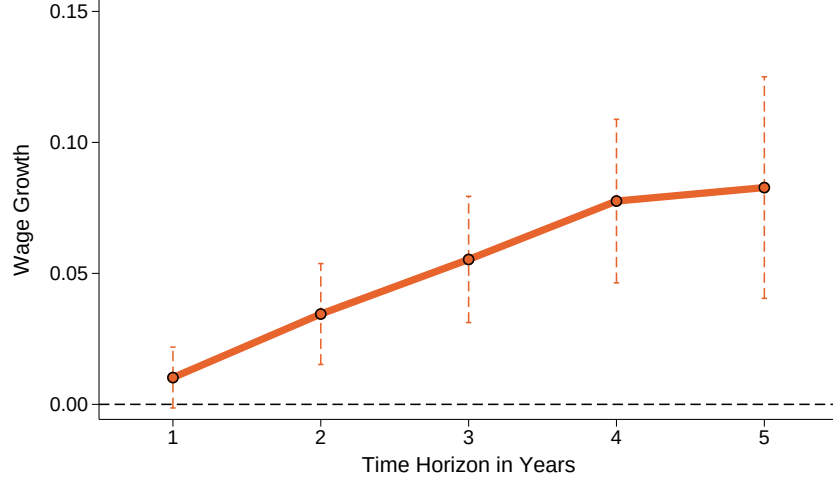
$$\Delta \log w_{i,t+h} = \beta_h \text{Modern}_{i,t} + \Gamma X_{i,t} + \alpha_t + \alpha_i + \zeta_{i,t}, \quad (14)$$

where $\Delta \log w_{i,t+h}$ is the change in worker i 's log wage between years t and $t+h$. The vector $X_{i,t}$ includes the same worker characteristics as in (13), together with wage-decile fixed effects. Moreover, α_t and α_i denote time and worker fixed effects, respectively. The parameter of interest is β_h , which captures the average difference in wage growth over h years between modern and traditional workers, conditional on observable characteristics and fixed effects.

Figure 3 presents the estimated future wage premium associated with working in the modern sector across different time horizons. The premium is statistically insignificant at short horizons but increases over time, reaching its peak after four years. At this horizon, modern workers experience wage growth that is approximately 4% higher than that of traditional-sector workers. Crucially, because the specification includes worker fixed effects, identification comes from workers who change sectoral status. This allows us to compare the wage growth trajectories of the same worker during periods of modern and traditional employment and to control for the potential sorting of workers with higher learning ability into the modern sector. Hence, the results suggest that workers who switch from the traditional sector to the modern sector experience cumulative wage gains in subsequent years.

We do not condition on workers remaining in the modern sector in subsequent years. This is

Figure 4: Peer Effects on Wage Growth



Notes: Figure displays the effect of peer wages on workers’ future wage growth based on the estimates of β_h from equation (15). Peers are defined as all other workers in the same Region $\times$ Year $\times$ Industry $\times$ Occupation $\times$ Firm-Size Bracket. Each dot represents the estimated effect at the time horizon $t+h$. Controls include age, education, gender, occupation, industry, region, experience, wage decile, and time fixed effects. Vertical dotted lines denote 95 percent confidence intervals.

consistent with our model, in which human capital is portable across sectors. As a result, our estimates should be interpreted as a lower bound on the dynamic modern-sector premium that we would obtain if we conditioned on workers who remain in the modern sector. In addition, the estimated effects become statistically insignificant after five years. This is natural in our setting, since workers may switch sectors over time in response to idiosyncratic skill shocks.

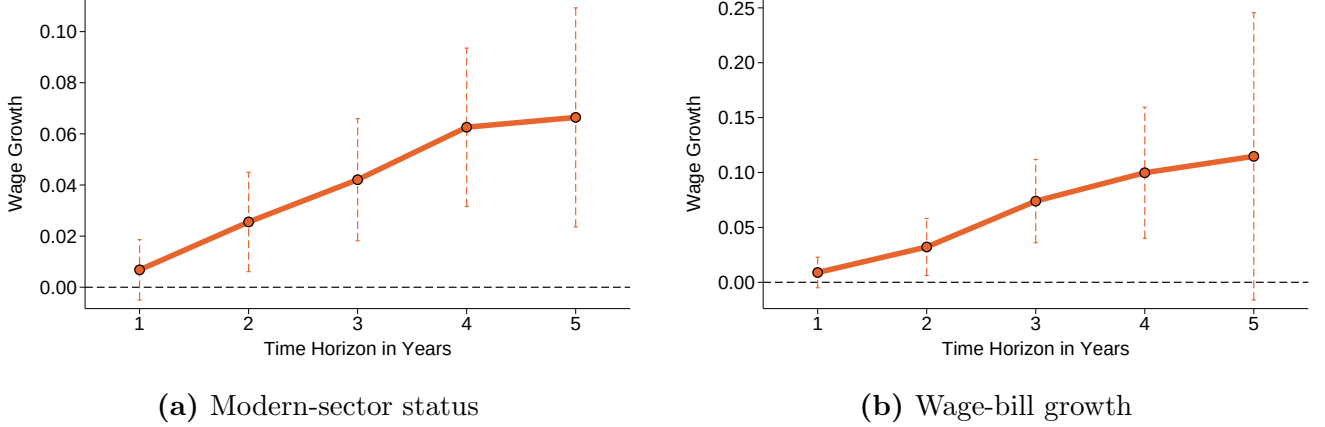
Which forces drive the dynamic modern-sector wage premium? Following Jarosch et al. (2021) and Herkenhoff et al. (2024), we examine whether interactions with better-paid peers contribute to workers’ wage growth. In our framework, workers interact with others in the same labor market, and access to more skilled peers accelerates human capital accumulation. Accordingly, we define a labor market in the data as a region $\times$ industry $\times$ occupation $\times$ firm-size bracket cell in a given year. For each worker i , we then compute the average wage of all other workers in the same labor market, denoted by $w_{-i,t}$, as a proxy for peer skill. We estimate the following reduced-form equation:

$$\Delta \log w_{i,t+h} = \beta_h \log w_{-i,t} + \Gamma X_{i,t} + \zeta_{i,t}, \quad (15)$$

where β_h captures how exposure to better-paid peers affects a worker’s future wage growth, and $X_{i,t}$ contains the same worker-specific controls as in our previous specifications.

Figure 4 presents the estimated effects of peer quality on wage growth across different time horizons. We find strong evidence that workers exposed to better-paid peers experience significantly higher future wage growth. This effect increases with the time horizon, suggesting that peer quality has a cumulative impact on skill accumulation. In terms of magnitude, a one-standard-deviation increase in the average wage of a worker’s peers is associated with wages that are approximately 4% higher five years later. These findings align with the literature on learning from coworkers (Jarosch et al., 2021; Herkenhoff et al., 2024) and learning in cities (Crews, 2024; Lhuillier, 2024).

Figure 5: Peer Effects on Wage Growth, Robustness



Notes: The figure reports estimates of β_h from equation (15) under two robustness specifications. In Panel 5(a) we control for an indicator for modern-sector employment, Modern_{it} . In Panel 5(b) we control for the growth of the labor market’s total wage bill between $t-2$ and $t-1$, $t-1$ and t , and t and $t+1$. The remaining controls are the same as in Figure 4. Vertical dotted lines denote 95 percent confidence intervals.

They also provide suggestive evidence that the quality of workers’ interactions plays an important role in shaping their wage trajectories.

It is natural to consider alternative mechanisms that could drive the dynamic modern-sector premium. Our data allow us to examine whether the peer-learning channel is confounded by two other forces. First, sectors may differ in characteristics other than peer composition that affect wage growth. For instance, consistent with the results of Ma et al. (2024), modern-sector workers may have greater access to on-the-job training, leading to faster human capital accumulation than traditional-sector workers. To assess whether our peer-learning estimates are confounded by other sector-specific channels, such as training, we estimate a variant of (15) that directly controls for workers’ modern-sector status.¹⁷ Panel 5(a) reports the estimates of β_h from this specification. The estimates remain similar to those in Figure 4, suggesting that the peer-learning channel is not driven solely by other sector characteristics that affect human capital accumulation.

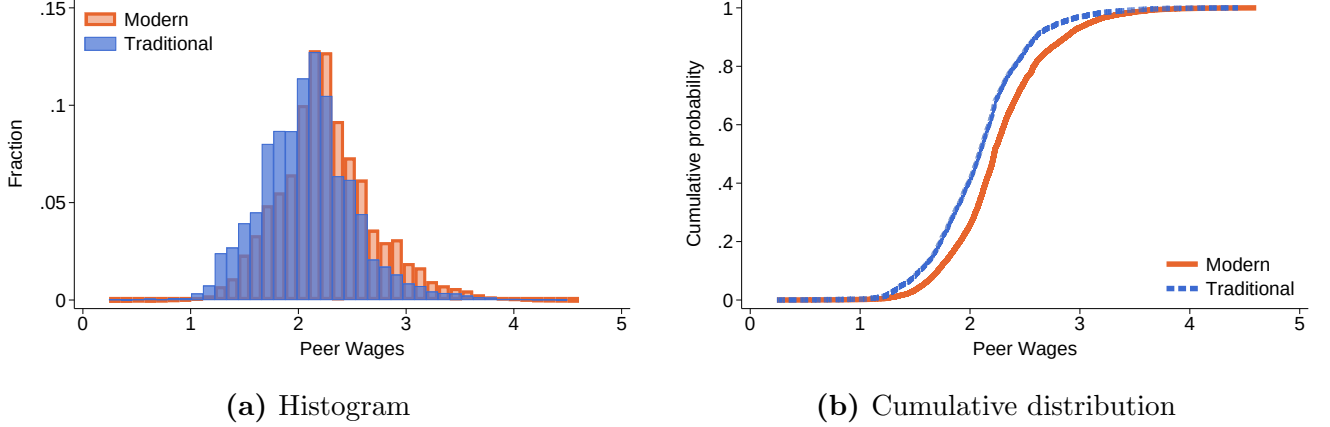
Second, an alternative mechanism is that the estimated effects reflect labor-market-level productivity gains that are passed on to workers rather than learning from peers. Following Jarosch et al. (2021), we estimate a version of (15) that controls for growth in the labor market’s total wage bill between $t-2$ and $t-1$, $t-1$ and t , and t and $t+1$.¹⁸ Figure 5(b) displays the estimates, which remain positive and increase with the horizon, suggesting that the results are not driven by common shocks to workers’ labor markets. Although the standard errors increase at longer horizons, the point estimates are even larger in magnitude.

Having established that peers matter for wage growth, we now examine whether modern-sector workers systematically have access to better peers than traditional-sector workers. Figure 6 compares the distribution of peer wages, $w_{-i,t}$, for workers in the two sectors.

¹⁷Perhaps surprisingly, participation in formal training programs appears rare in our data. Only 2% of job spells report participation in an employment or training program.

¹⁸Recall that our data do not allow us to identify the establishments or firms in which workers are employed.

Figure 6: Distribution of Peer Wages for Modern and Traditional Workers



Notes: Figure 6 displays the distribution of log peer wages, $\log w_{-i,t}$, for modern and traditional workers. Panel 6(a) shows the histogram and panel 6(b) the empirical cumulative distribution function. Peers are defined as workers in the same Region $\times$ Year $\times$ Industry $\times$ Occupation $\times$ Firm-Size Bracket.

The figure reveals that modern workers indeed have better-paid peers. On average, the peers of modern workers earn wages approximately 20% higher than the peers of traditional workers. In addition, the distribution of peer wages for modern workers first-order stochastically dominates that of traditional workers. Because peer wages average over all workers in a labor market, the two distributions naturally overlap: modern and traditional workers who share the same labor market face similar peer groups. The dominance of the modern-sector distribution therefore reflects the sorting of modern workers into labor markets with better-paid peers. This finding provides the missing link in our narrative: modern workers experience higher wage growth partly because they interact with better-paid peers, and these interactions contribute to faster skill accumulation.

We note that this evidence captures differences in peer exposure across labor markets rather than learning segmentation within them. It is rare to observe workers' interactions directly within a labor market, and our data are no exception. Our model addresses this limitation by allowing the frequency of interactions between modern and traditional workers within a labor market to differ. We estimate this learning segmentation in our quantitative analysis and study its implications.

Finally, we provide suggestive evidence supporting our modeling choice for the learning technology. Equation (3) assumes that workers can learn from peers across the entire skill distribution within a labor market, from the least to the most skilled. One way to assess this assumption is to compare the effects of better-paid peers for workers at the top and bottom of the wage distribution. If workers learned only from more skilled peers, we would expect stronger peer effects among low-wage workers, who have a larger pool of more skilled workers from whom to learn. Conversely, if workers learned only from less skilled peers, we would expect stronger effects among high-wage workers. To explore this asymmetry, we estimate the following specification:

$$\Delta \log w_{i,t+h} = \beta_h^+ \text{Above}_{i,t} \times \log w_{-i,t} + \beta_h^- \text{Below}_{i,t} \times \log w_{-i,t} + \Gamma X_{i,t} + \zeta_{i,t}, \quad (16)$$

where $\text{Above}_{i,t}$ and $\text{Below}_{i,t}$ are indicator variables equal to one when worker i is above or below

the mean wage of her peers, respectively.¹⁹

Table D2 in Appendix D reports the estimates of β_h^+ and β_h^- across different time horizons. Both groups display positive and sustained responses to peer quality at every horizon. While the point estimates are larger for workers above the mean, both sets of coefficients are economically and statistically significant, and they converge at the longest horizons. We interpret this as suggestive evidence that workers learn from the full pool of peers in their labor market rather than only from a subset of them.

3.2.3 Taking Stock

In this section, we provided evidence for the model’s main predictions and the mechanism connecting them. Modern-sector workers earn higher wages throughout their careers, and a large share of this wage gap reflects the sorting of higher-skilled workers into the modern sector. We also documented a dynamic modern-sector wage premium, with modern workers experiencing faster future wage growth than traditional workers. Consistent with the model, exposure to better-paid peers predicts higher future wage growth, and modern-sector workers are systematically exposed to such peers. These sorting patterns therefore create better learning opportunities in the modern sector, leading to faster human capital accumulation and steeper wage trajectories, while wages in the traditional sector remain relatively flat over the life cycle.

It is important to recognize that peer learning is not the only force shaping human capital accumulation across the modern and traditional sectors in developing economies. Although our peer-effect estimates remain robust after controlling for modern-sector status, more detailed information on firm-provided training, such as that used by Ma et al. (2024), would be necessary to quantify the contribution of on-the-job training to the wage trajectories we document. In addition, consistent with the evidence in Donovan et al. (2023), traditional-sector workers may face riskier jobs, which could also limit wage growth over the life cycle. Our model and empirical analysis abstract from unemployment and these broader labor-market flows.

Despite these limitations, our findings provide consistent support for worker sorting and peer learning as important forces shaping wage trajectories across sectors. In the next section, we quantify the model and use it to assess the aggregate importance of peer learning and learning segmentation for income and welfare.

4 Model Quantification

This section describes the estimation of the model parameters. We summarize the parameters described in our theory and discuss the various strategies employed for their estimation. Next, we examine the variation in the data that helps identify these parameters.

¹⁹We also include the non-interacted indicator $\text{Above}_{i,t}$ in the vector of controls $X_{i,t}$.

4.1 Set Up

We define a labor market in our data as a Region–Industry–Occupation triplet, consistent with the empirical exercises in Section 3. The EPS records information for 13 regions, 10 industries, and 10 occupations, yielding a total of $N = 1,300$ labor markets.

To take the model to the data, we specify two functional forms. First, we assume the initial skill distribution, $\psi(z)$, is Pareto, with the location parameter normalized to one and shape parameter ϑ . Second, we parameterize the production technology for traditional-sector workers as $\phi_n(z) = \phi(z)z^\eta$, with $\eta < 1$.

Since time is continuous in our model, we interpret an interval $[t, t + 1)$ as one year. Furthermore, we average wages and transition probabilities across modern and traditional sectors over 2002–2016 and interpret these averages as the steady state of the economy.

4.2 Estimation

With our functional form assumptions, we have 13 parameters

$$\{\rho, \tau, \nu, \vartheta, \mu, \delta, \theta, \gamma, \pi^M, \pi^T, \sigma^M, \sigma^T, \eta\},$$

in addition to the labor-market productivity, $\{A_n\}_{n=1}^N$, and bilateral migration costs, $\{\xi_{nm}\}_{n,m=1}^N$. We divide these parameters into three groups, according to the estimation strategy we follow to estimate each of them.

External Calibration (3 parameters): ρ, τ, ν

We set workers' discount rate, ρ , to 0.05, a standard value in the literature (Akçigit et al., 2021). We set payroll taxes in the modern sector to $\tau = 0.18$, consistent with Chilean statutory regulations. These costs include pension and social security contributions. Finally, we set the inverse of worker's dispersion parameter of labor-market idiosyncratic preference shocks to $\nu = 1/1.5$, following Traiberman (2019).

Given our value for ν , we estimate bilateral migration costs, $\{\xi_{nm}\}_{n,m=1}^N$, non-parametrically using the bilateral migration flows across labor markets observed in the EPS. To conduct this nonparametric estimation, we impose symmetric migration costs, as we explain in Appendix E.

Minimum Distance Estimated Parameters (3 parameters): δ, μ, ϑ

We estimate the death shock, δ , using the implied age distribution. Letting $H(a)$ denote the age CDF, the model implies the log-linear relation

$$\log(1 - H(a)) = -\delta a.$$

We therefore estimate δ by regressing $\log(1 - H^e(a))$ on age a without a constant, where $H^e(a)$ is the empirical age CDF.

For the migration rate, μ , we use the worker panel data and set it equal to the overall fraction of workers who change labor markets in a given year. In the model, migration opportunities arrive according to a Poisson process with intensity μ , implying that a fraction μ of workers receive an opportunity to move each period. Using the EPS, we identify movers and non-movers and compute μ as the fraction of workers who move across labor markets between consecutive survey waves.

We estimate the Pareto shape parameter ϑ directly from the cross-section of entry wages. Formal wages are proportional to skill, $w_n^M(z) = A_n z / (1 + \tau)$, so within each labor market the wages of newborns inherit the Pareto tail index ϑ up to the local scale $A_n / (1 + \tau)$. We identify newborns as formal workers aged at most 25 with fewer than two years of experience, take each worker's first observed wage, rank workers by wage within each market, and keep the top quartile. We then run the rank-size regression

$$\log(\text{rank}_{i,n} - \tfrac{1}{2}) = c_n - \vartheta \log w_{i,n} + \zeta_{i,n},$$

where the market fixed effect c_n absorbs the local wage scale and the slope on log wages identifies ϑ . The $-\frac{1}{2}$ rank correction removes the small-sample bias of the OLS tail-exponent estimator (Gabaix and Ibragimov, 2011).

Our estimation delivers $\delta = 0.037$, $\mu = 0.08$ and $\vartheta = 2.54$. The estimated value of δ implies that around 3.7% of workers in our model exit employment each year and that, at any point in time, active workers have a remaining expected work life of about 26 years. This number is higher than Chile's overall mortality rate, which is natural since δ captures not only death but also exits from employment into unemployment or out of the labor force. Moreover, our estimated value of μ lies in the lower range of the values estimated by Crews (2024) for the United States.

Simulated Method of Moments (7 parameters): $\eta, \theta, \gamma, \pi^M, \pi^T, \sigma^M, \sigma^T$

We estimate the remaining 7 parameters using a Simulated Method of Moments approach, jointly with an inversion procedure to recover the market-specific productivity shifters $\{A_n\}_{n=1}^N$.

The approach to estimate market productivity is as follows. Given the estimates of the previous parameters, and a guess for the remaining ones, we recover the vector of market productivity shifters by exactly matching average wages in each labor market. Formally, the wage equation (1) implies that

$$A_n = \frac{\bar{w}_n}{s_n^M \frac{\mathbb{E}_n[z|M]}{1+\tau} + s_n^T \mathbb{E}_n[z^\eta|T]}, \quad (17)$$

where $\bar{w}_n$ is average wage in market n , s_n^M and s_n^T are the shares of modern and traditional workers in market n , respectively, while $\mathbb{E}_n[z|M]$ and $\mathbb{E}_n[z^\eta|T]$ are the conditional average marginal product of modern and traditional workers in market n . We use data on average wages per labor market to construct the numerator of the RHS of equation (17). For the denominator, in contrast, we need to solve the model given an initial vector of productivity and parameter to construct the

equilibrium objects entering this expression. Therefore, we implement an iterative procedure until convergence of the vector $\{A_n\}_{n=1}^N$ given parameters. We nest this inversion procedure into our SMM estimation routine described below.

Turning to our SMM estimation, we estimate the seven structural parameters $\eta, \theta, \gamma, \pi^M, \pi^T, \sigma^M, \sigma^T$ using ten data moments. Four are aggregate cross-sectional moments: the modern-sector employment share S^M , the one-year sector transition probabilities P_{MM}^{1y} and P_{TT}^{1y} , and the variance of log wages within the modern sector $\text{Var}(\log w \mid M)$. These four aggregate moments are reported in Table 1. The remaining six moments are the diagonal of the within-sector wage-quintile transition probabilities at a five-year horizon, $P_{qq|M}^{5y}$ and $P_{qq|T}^{5y}$ for $q \in \{2, 3, 4\}$. We define the quintiles of the sector-specific wage distribution and restrict workers to be observed in the same sector at both endpoints. We exclude the bottom and top diagonals in each sector, since the smooth-Brownian specification lacks structural mechanisms that drive the persistence patterns the data shows at the extremes of the wage distribution. Furthermore, we compute the sector and the wage-quintile transition probabilities deliberately at different horizons. The one year sector transitions emphasize diffusion crossings of the sector cutoff and so isolate σ^S , while five-year within-sector quintile persistence emphasizes cumulative drift along workers' careers.

The mapping from moments to parameters has a clean structure. The within-sector quintile diagonals identify the sector-specific drift $\mu^S = \gamma Z^S$. Persistence at higher quintiles is informative about μ^M , while persistence at lower quintiles is informative about μ^T ; the asymmetry in within-sector persistence patterns separates the two. Holding μ^S fixed by these moments, the one-year sector-transition probabilities P_{MM}^{1y} and P_{TT}^{1y} identify σ^M and σ^T via the rate of diffusion-induced crossings of the sector cutoff. [Perla et al. \(2021\)](#) adopt a similar strategy to estimate parameters governing the law of motion of firm productivity when it follows a geometric Brownian motion, as in our setting. The modern share S^M and the variance of log wages $\text{Var}(\log w \mid M)$ jointly pin down the technology curvature η and the level structure of the steady-state distribution. Finally, the learning-segmentation parameters π^M and π^T are identified by the difference between own-sector and cross-sector contributions to the vibrancy Z^S : holding γ and the wage distributions fixed, the implied drift in each sector, and hence the within-sector quintile persistence, depends on how much of Z^S comes from own-sector versus cross-sector composition.

Overall, we have a total of 10 moments that we use to identify 7 parameters. As the model is over-identified, in current work we conduct robustness exercises by varying the moment set and assessing the sensitivity of the estimates to alternative weighting matrices.

Formally, let $\Theta = (\eta, \theta, \gamma, \pi^M, \pi^T, \sigma^M, \sigma^T)$ be the vector of parameters to estimate. We implement the SMM by minimizing the squared percent distance between the model-simulated moments, $\Psi^m(\Theta)$, and their empirical counterparts, Ψ^d :

$$\min_{\Theta} \sum_{i=1}^{10} \left(\frac{\Psi_i^m(\Theta) - \Psi_i^d}{0.5 (\Psi_i^m(\Theta) + \Psi_i^d)} \right)^2.$$

We use the TikTak algorithm for global optimization of [Arnoud et al. \(2019\)](#) to search over the pa-

parameter space.²⁰ At each iteration of the optimization routine, we invert the equilibrium mapping, (17), to non-parametrically recover $\{A_n\}_{n=1}^N$.

Table 2 reports the estimates. Panel A reports the values of the moments in the data and in the model. Overall, the model performs well in fitting the 10 moments with 7 parameters. For the aggregate moments, the model replicates three of them with high precision: the one-year sector transition probabilities and the variance of log wages in the modern sector. Nevertheless, the model does not accurately replicate the share of modern workers in the economy. In the estimated model, there are too few modern-sector workers. Despite the simple and parsimonious law of motion for skills in both sectors, the model does a good job matching the sector-specific transition probabilities across wage quintiles.

Panel B reports the estimated SMM parameters. We find strong diminishing returns in the traditional sector, with η significantly below one. We also estimate the power-mean parameter governing vibrancy in each sector to be 1.67. This value is somewhat lower than the estimate in Crews (2024) for the United States. The discrepancy likely reflects the fact that Crews (2024) focuses on interactions within cities and incorporates dynamic agglomeration forces, while we focus on interactions within a narrower definition of labor markets and do not include agglomeration forces.

Moreover, we find strong evidence of learning segmentation in both sectors. Relative to a random-interactions benchmark with $\pi^M = \pi^T = 0.5$, the estimated learning weights imply that modern workers are 3 times more likely to learn from a modern peer than from a traditional peer. Similarly, traditional workers are almost 4 times more likely to learn from peers in their own sector than from peers in the modern sector. Finally, we find very similar volatility of worker skills across sectors, with modern workers having slightly more volatile skills.

4.3 Decentralized Equilibrium

With our estimated parameters, we briefly illustrate the decentralized equilibrium. To make estimation feasible, we aggregate the initial 1,300 labor markets into 8 market clusters, grouping regions by geographic proximity and industries and occupations by wage.

First, Figure 7 displays the mass of workers across markets and sectors. We highlight three features of the baseline distribution. First, more productive markets, in terms of TFP, naturally attract more workers. Second, we observe positive sorting, with more productive workers selecting into the modern sector. The vertical dotted lines in the figure denote the modern-sector skill threshold, which generally decreases with local TFP. This implies that more productive markets typically have a higher share of modern-sector workers. However, this relationship is not perfect, since markets also differ in migration costs. Finally, we observe some bunching of workers in the traditional sector just below the threshold. This pattern is due to the fact that the estimated volatility of skills is higher in the modern sector than in the traditional sector. Hence, modern-sector workers just above the cutoff move through the skill distribution faster than traditional-sector workers just below the threshold.

²⁰We use a Sobol sequence of 5000 starting points and a simplex search method for local optimization using the best 50 initial candidates.

Table 2: SMM estimation results
Panel A: Moments

Moment Description	Data	Model
Modern-to-modern transition (1 year)	0.97	0.95
Traditional-to-traditional transition (1 year)	0.92	0.92
Modern-sector employment share	0.69	0.60
Variance of log wages, modern sector	0.42	0.42
Modern $q_2 \rightarrow q_2$ transition (5 years)	0.31	0.33
Modern $q_3 \rightarrow q_3$ transition (5 years)	0.37	0.38
Modern $q_4 \rightarrow q_4$ transition (5 years)	0.48	0.52
Traditional $q_2 \rightarrow q_2$ transition (5 years)	0.37	0.41
Traditional $q_3 \rightarrow q_3$ transition (5 years)	0.33	0.33
Traditional $q_4 \rightarrow q_4$ transition (5 years)	0.39	0.34

Panel B: Estimates

Parameter	Description	Estimate
η	Decreasing returns, traditional	0.536
θ	Power-mean, vibrancy	1.673
γ	Peer-learning intensity	0.0012
π^M	Learning segmentation, modern	0.766
π^T	Learning segmentation, traditional	0.794
σ^M	Volatility of log skill, modern	0.128
σ^T	Volatility of log skill, traditional	0.110

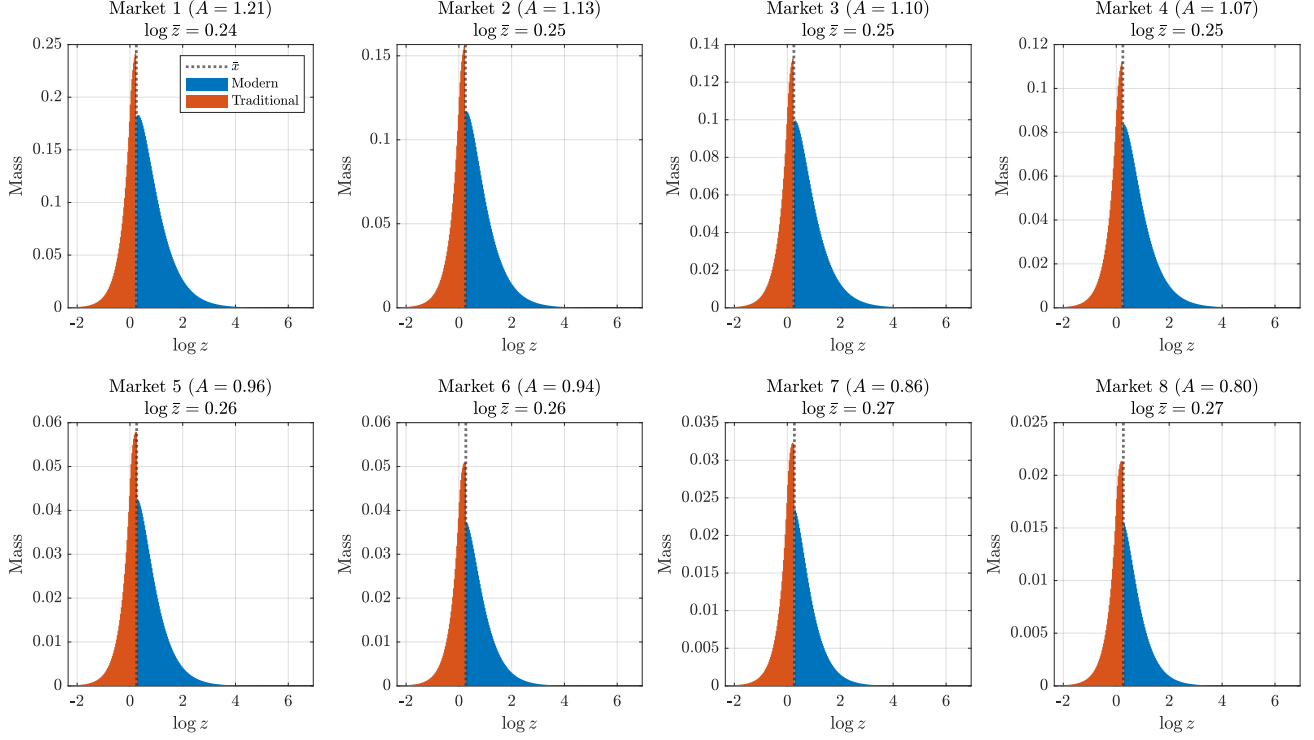
Notes: Panel A displays the empirical moments computed from the EPS and their model counterparts at the estimated parameters. Panel B reports the SMM parameter estimates obtained by minimizing the weighted distance between the model and data moments.

Second, Figure 8 displays the estimated average skill growth across the economy. The figure shows the drift in (3) across markets and sectors, normalized by the smallest drift, which corresponds to the traditional sector in the first market. We observe large heterogeneity in mean skill growth rates across markets and sectors. Heterogeneity across sectors is especially pronounced: modern-sector workers in all markets accumulate skills faster than traditional-sector workers in any market. Similarly, there is substantial heterogeneity within markets, with modern workers experiencing a human capital growth rate that is almost twice that of traditional workers in the same market. Overall, modern-sector workers in the market with the fastest human capital accumulation accumulate skills at a rate that is almost three times that of traditional-sector workers in the market with the lowest rate of skill accumulation.

5 The Role of Learning in Economic Development

With the estimated model, we conduct two counterfactual exercises to explore the role of peer learning in economic development in the Chilean context. Our first exercise studies what happens to aggregate income and welfare when the cost of operating in the modern sector decreases. In the model, modern-sector costs are payroll taxes, τ , which are passed on to workers in the form of lower

Figure 7: Worker's Mass Across Markets and Sectors



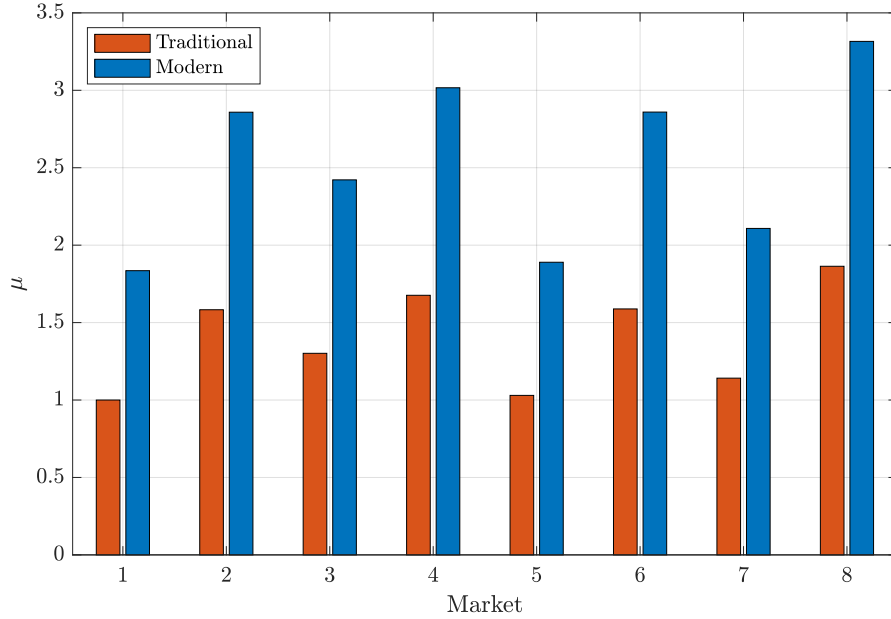
Notes: Figure 7 displays workers' mass across markets and sectors for 8 aggregate markets clustered by region, industry, and occupation. The x-axis shows log skill. Vertical dotted lines indicate the modern-sector skill cutoff in each market. Markets are ordered by TFP, A .

wages. We therefore consider reductions of 5, 10, and 18 p.p. in τ , with the last case corresponding to the elimination of employers' social security contributions. This exercise is informative about the role of learning in development because higher-skill workers sort into the modern sector. Reducing the cost of operating in this sector increases its static payoff through higher wages, but it can also improve learning opportunities for workers entering the modern sector by exposing them to higher-skill peers.

Our second exercise focuses directly on barriers to learning for traditional-sector workers. We study the impact on aggregate income and welfare if traditional-sector workers had the same learning opportunities as modern-sector workers. We do this by setting $\pi^T = 1 - \pi^M$, which implies that traditional-sector workers meet higher-skilled workers at the same frequency as modern-sector workers do. In contrast to the first exercise, this is not a policy counterfactual. The model does not specify a structural relationship between the learning weights and policy instruments that could change them. Nevertheless, we view this exercise as important because it provides a reduced-form measure of the aggregate importance of learning barriers faced by traditional-sector workers.

Do we expect learning segmentation to always be a barrier to economic development? The answer, perhaps surprisingly, is no. Workers impose a learning externality on others. When they choose labor markets and sectors, they do not internalize that these decisions affect other workers' learning opportunities. Eliminating learning segmentation altogether can actually reduce the

Figure 8: Learning Intensity Across Markets and Sectors



Notes: Figure 8 displays the drift in the law of motion for workers' skill, (3), in the estimated stationary equilibrium. The x-axis denotes 8 aggregate markets clustered by region, industry, and occupation; colored bars denote sectors.

learning opportunities of the most skilled workers, because they are now exposed to less skilled peers, creating a crowding-out effect. At the same time, eliminating segmentation increases the learning opportunities of traditional-sector workers, who become more exposed to higher-skilled counterparts. Whether, in the aggregate, removing learning segmentation raises or lowers income and welfare is therefore a quantitative question. Importantly, this crowding-out effect also appears in semi-endogenous growth models such as Buera and Oberfield (2016), who show that in a world with free trade, idea diffusion can reduce growth in some countries.

5.1 Reducing Cost of Operating in the Modern Sector

Table 3 displays the aggregate effects of reducing the cost of operating in the modern sector. The first and second columns report the payroll tax and the share of modern workers in the overall economy, respectively. Lower payroll taxes increase the share of modern workers in the economy. Interestingly, given market productivities and the curvature of the traditional-sector production technology, not all workers choose to operate in the modern sector even when payroll taxes are completely eliminated.

Column 4 of Table 3 shows the change in income per capita. Lowering the cost of operating in the modern sector slightly increases aggregate income, from 0.07% to 0.42%. The change in income per capita is driven by two forces. First, a static force raises income for workers already in the modern sector, who experience an increase in wages. Second, a dynamic force raises income for workers who switch sectors, as they accumulate more human capital after the policy change because they are exposed to more skilled peers. In ongoing work, we are decomposing the effects on output into

Table 3: Counterfactual Results

(1)	(2)	(3)	(4)	(5)
τ	Share Modern	ΔY	Δ Sector TFP Gap	Δ Welfare Learning
0.18	0.59	...	...	...
0.13	0.65	0.07	-3.20	0.22
0.08	0.71	0.11	-5.93	0.47
0.00	0.80	0.42	-7.42	1.00

Notes: Table 3 displays changes in aggregate objects in counterfactual economies with different payroll tax rates, τ , relative to the baseline stationary equilibrium. Y denotes income per capita; the sector TFP gap is the difference between modern- and traditional-sector labor productivity; and “Welfare Learning” denotes workers’ continuation value from learning in (7). All changes are reported in percentage terms.

these two distinct channels.

Productivity gaps between modern- and traditional-sector workers also decrease when payroll taxes are lowered. Column 5 of Table 3 shows the change in the labor productivity gap between the modern and traditional sectors. Interestingly, TFP decreases in both sectors because workers who move across sectors are less productive than the baseline marginal worker. However, the decline in TFP is less pronounced in the traditional sector, leading to a smaller average sectoral productivity gap across markets.

Finally, the last column of Table 3 informs the welfare gains from learning generated by the policy. The column reports the change in the continuation value from learning, which corresponds to the second term in the worker’s value function in (7). Importantly, we find that, in the aggregate, workers gain from the reduced cost of working in the modern sector. In consumption-equivalent terms, the direct welfare gains from improved learning opportunities reach 1% when payroll taxes are completely eliminated. These gains arise because lower payroll taxes increase the share of workers in the modern sector, where learning opportunities are better on average.

However, this aggregate number masks substantial heterogeneity across workers. As discussed above, changes in sectoral composition affect not only movers, who may gain access to better learning opportunities, but also incumbent workers, whose peer composition changes after the policy. Some workers benefit from being exposed to more skilled peers, while others may experience worse learning opportunities as lower-skill workers enter their sector or labor market. Therefore, the welfare gains in the last column average over workers who gain from the policy and workers whose learning opportunities deteriorate. This distinction is important because learning is an externality: workers do not internalize how their sectoral and market choices affect the human capital accumulation of others. We decompose the winners and losers from the policy below.

Figure 9 decomposes the welfare gains from learning when payroll taxes are completely eliminated for workers with different skill levels. The x-axis denotes workers’ log skill, while the y-axis denotes the consumption-equivalent welfare gains from the learning channel. The vertical dashed

line denotes the average modern-sector cutoff across markets in the baseline equilibrium, while the vertical solid line denotes the average modern-sector cutoff across markets when payroll taxes are eliminated.

The policy has heterogeneous effects on workers depending on their skill. First, for workers who are always in the modern sector, the policy tends to deteriorate learning opportunities. The reason is that, after payroll taxes are eliminated, they are exposed to less skilled workers who move into the modern sector. This generates welfare losses of around 1% for the most skilled workers in the economy. However, the effect is not negative for all always-modern workers, since the learning technology allows workers to also learn from lower-skilled peers. In fact, always-modern workers close to the initial cutoff experience welfare gains.

Second, the largest welfare gains from the policy are concentrated among movers: workers who switch from the traditional sector to the modern sector after the policy change. This group corresponds to workers located between the baseline modern-sector cutoff and the new cutoff. By moving into the modern sector, these workers interact with higher-skilled peers and experience an improvement in their learning opportunities that reaches more than 3%.

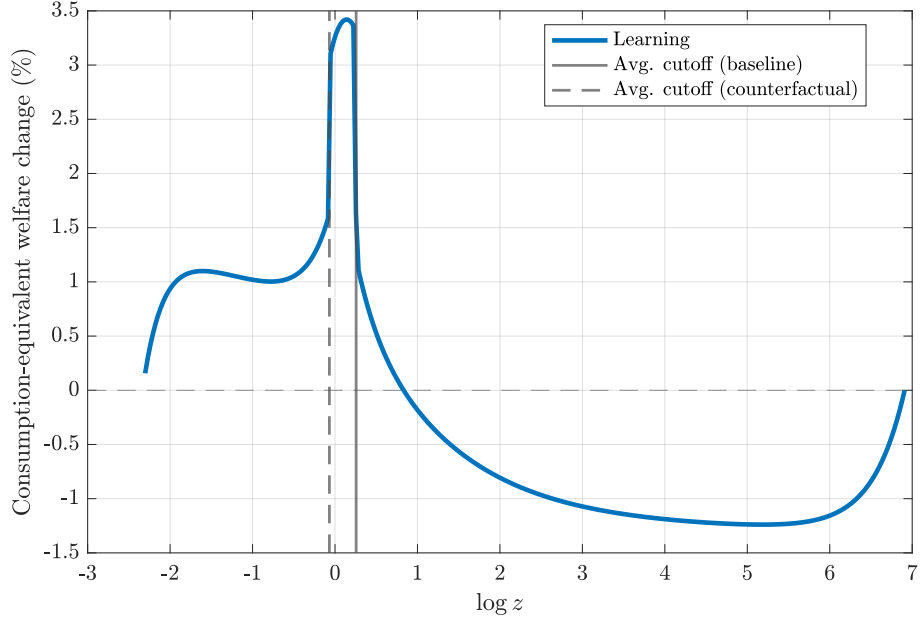
Finally, always-traditional workers, those who begin and remain in the traditional sector after the policy, also experience welfare gains through learning. The reason is that these workers still interact with modern-sector workers, even in the presence of learning segmentation. Since the policy increases the mass of workers in the modern sector, and since these new modern-sector workers benefit from improved learning opportunities, the option value of remaining in the traditional sector also increases. For workers who remain traditional, it is now more likely to meet high-skill workers in the economy because the modern sector is larger. On average, always-traditional workers experience a welfare gain of 1.2%, movers experience welfare gains of 3.2%, and always-modern workers slightly gain on net, with welfare gains of 0.03%.

5.2 Improving Learning Opportunities for Traditional Workers

We now turn to the second counterfactual exercise, which focuses on the effects of improving the learning opportunities of traditional-sector workers. The learning technology, governed by (5), (4), and (3), together with the estimated wedges, π^M and π^T , implies that it is relatively harder for traditional-sector workers to meet high-skill workers in the economy than it is for modern-sector workers. Interactions in both sectors are highly biased and, since high-skill workers sort into the modern sector, traditional-sector workers tend to interact mostly with other low-skill traditional-sector workers. Hence, we consider an exercise in which traditional-sector workers interact with high-skill workers at the same rate as modern-sector workers do. In our setting, this corresponds to setting $\pi^T = 1 - \pi^M$.

Improving learning opportunities for traditional-sector workers also leads to sizable aggregate effects. The exercise generates welfare gains slightly above 0.7%. Interestingly, the share of modern workers slightly decreases to 0.57. This result is a consequence of improving the value of operating in the traditional sector. Marginal workers who operate in the modern sector because of its better learning environment now switch to the traditional sector, where learning opportunities are the

Figure 9: Welfare Gains from Peer Learning - No Payroll Taxes



Notes: Figure 9 decomposes learning welfare gains in the counterfactual economy with no payroll taxes, $\tau = 0$. The x-axis denotes log skill, while the y-axis denotes consumption-equivalent welfare gains. The vertical dashed line denotes the average modern-sector cutoff in the baseline economy. The vertical solid line denotes the average modern-sector cutoff in the economy with no payroll taxes.

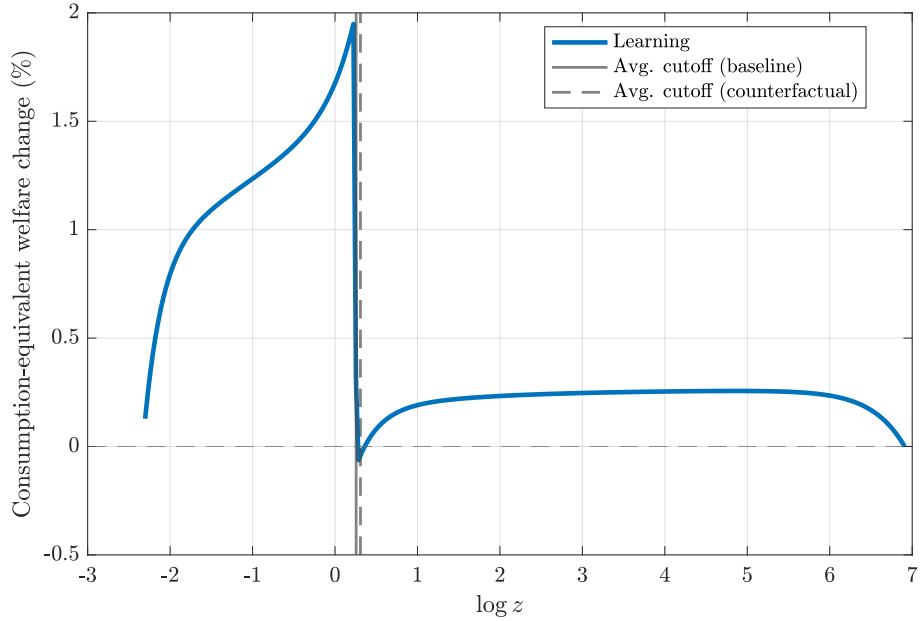
same and they can earn a higher wage. These results point to the fact that the traditional sector is not always a stepping stone in which workers accumulate human capital before switching to the modern sector.

Figure 10 decomposes the welfare gains for workers with different skill levels. Lower-skill workers who start in the traditional sector experience welfare gains of around 1.5% as a consequence of the improved learning opportunities. Interestingly, contrary to our previous exercise, modern-sector workers do not experience welfare losses and are generally better off in the new stationary equilibrium. The reason is that some workers switch from the modern sector to the traditional sector after learning opportunities in the traditional sector improve. This increases the average skill of workers who remain in the modern sector, improving the learning opportunities of always-modern workers.

6 Conclusion

This paper has studied how segmented learning markets shape human capital accumulation and economic development in dual economies. We developed a quantitative model in which workers sort across labor markets and between a modern and a traditional sector, accumulating skills by learning from peers within their own sector. The model predicts that higher-skill workers select into the modern sector and that modern-sector workers enjoy faster wage growth, a dynamic premium we document in Chilean panel data and trace to learning from better-paid peers. Estimating

Figure 10: Welfare Gains from Peer Learning - $\pi^T = 1 - \pi^M$



Notes: Figure 9 decomposes learning welfare gains in the counterfactual economy with no payroll taxes, $\tau = 0$. The x-axis denotes log skill, while the y-axis denotes consumption-equivalent welfare gains. The vertical dashed line denotes the average modern-sector cutoff in the baseline economy. The vertical solid line denotes the average modern-sector cutoff in the economy with no payroll taxes.

the model, we find strong learning segmentation: because high-skill workers concentrate in the modern sector and interactions occur largely within sectors, traditional-sector workers accumulate human capital far more slowly. Our counterfactuals show that lowering the cost of operating in the modern sector and improving learning opportunities for traditional-sector workers both raise aggregate income and welfare. Yet, because learning is a local externality that workers do not internalize, these gains are unevenly distributed and eliminating segmentation is not always beneficial. The central message of the paper is that the segmentation of learning, rather than the mere coexistence of a modern and a traditional sector, is a key channel through which dual economies depress human capital accumulation and aggregate income.

Our framework points to several avenues for future research. First, our counterfactuals hold the structural weights of the learning technology fixed; linking these weights to concrete policy instruments such as training programs or policy re-organizing production structures would permit a fuller welfare analysis of interventions that reshape who interacts with whom. Second, richer data on workers' tasks, coworkers, and training would sharpen the identification of the peer-learning channel and its heterogeneity across sectors.

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A Theory Appendix

This appendix collects the auxiliary results behind the equilibrium characterization of Section 2.5, discusses how the analysis extends beyond the functional form benchmark, and provides the proofs of Lemmas 2–5, Propositions 1–2, and Lemma 1.

A.1 Auxiliary Results

The first lemma establishes the two properties of the value function that the sorting argument requires.

Lemma 2 (Shape of the value function). *In any stationary equilibrium with finite values: (a) V_n^S and V_n are strictly increasing in z for every n and S ; and (b) under Assumption 1, V_n is convex in $\log z$.*

The second lemma connects sorting to learning environments. Say that the allocation in market n is *sorted at $\bar{z}_n$* if N_n^T places no mass above $\bar{z}_n$ and N_n^M places no mass below $\bar{z}_n$.

Lemma 3 (Sorting ranks learning environments). *Suppose the allocation in market n is sorted at some $\bar{z}_n$ and Assumption 2 holds. Then g_n^M first-order stochastically dominates g_n^T , and consequently $Z_n^M \geq Z_n^T$ for every θ . The inequalities are strict if both sectors are populated and $\pi^M + \pi^T > 1$.*

The third lemma is the sorting result at the core of Proposition 1: given learning environments that favor the modern sector and common volatilities, optimal labor supply takes a cutoff form.

Lemma 4 (Labor supply is a cutoff). *Consider a stationary equilibrium with finite values in which $Z_n^M \geq Z_n^T$ in market n , and suppose $\sigma_M = \sigma_T$. Let $\hat{z}$ denote the unique crossing point of the wage schedules (1), defined by $\hat{z}/(1+\tau) = \phi(\hat{z})$, which under Assumption 1 equals the static cutoff (11). Then:*

- (i) *For any strictly increasing u and any ϕ satisfying our maintained conditions, $S_n^*(z) = M$ for all $z \geq \hat{z}$: the modern sector always attracts the top of the skill distribution.*
- (ii) *Under Assumption 1, labor supply is a cutoff policy with an interior cutoff: there exists $\bar{z}_n \in (0, \hat{z}]$ such that*

$$S_n^*(z) = T \text{ for } z < \bar{z}_n, \quad S_n^*(z) = M \text{ for } z > \bar{z}_n.$$

Moreover, $\bar{z}_n < \hat{z}$ whenever $Z_n^M > Z_n^T$.

The last lemma defines the constant κ that appears in Assumption 3 and formalizes the role of condition (12): it converts the bound on the entry moment into an a priori cap on the vibrancies that any sorted allocation can generate.

Lemma 5 (Vibrancy cap). *Let $\sigma \equiv \max\{\sigma_M, \sigma_T\}$, define*

$$\kappa \equiv \frac{\underline{L}}{c_\pi}, \quad c_\pi \equiv \max_{S \in \{M, T\}} \frac{\max\{\pi^S, 1 - \pi^S\}}{\min\{\pi^S, 1 - \pi^S\}}, \quad (\text{A1})$$

where $\underline{L} > 0$ is a lower bound on stationary market populations,²¹ and let

$$Z_H^* \equiv \frac{\delta - \theta^2 \sigma^2 / 2}{(\theta + 1) \gamma}. \quad (\text{A2})$$

Under Assumption 3, every sorted stationary allocation whose learning drifts are bounded by γZ_H^* generates vibrancies bounded by Z_H^* : $Z_n^S \leq Z_H^*$ for all n and S .

Under own-sector bias, $\pi^M, \pi^T \geq 1/2$, the constant simplifies to $c_\pi = \bar{\pi}/(1 - \bar{\pi})$ with $\bar{\pi} \equiv \max\{\pi^M, \pi^T\}$, so κ shrinks as learning becomes more segmented: segmentation allows a sector to concentrate exposure on its own most skilled members, decoupling its learning environment from the market-wide skill pool, which requires a more conservative cap.

The next two remarks complement Lemma 4. The first states the matching conditions that the stationary distribution satisfies at the cutoff, which we impose in the numerical solution. The second discusses how the sorting result generalizes beyond Assumption 1.

Remark 1 (Matching conditions). Under the cutoff policy, the general KFE (8) reduces to a two-region system: $N_n^T(z) = N_n(z)$ for $z < \bar{z}_n$ and $N_n^M(z) = N_n(z)$ for $z > \bar{z}_n$, with the stationary distribution satisfying continuity of mass and of the flux at the cutoff, $N_n^T(\bar{z}_n^-) = N_n^M(\bar{z}_n^+)$ and $J_n^T(\bar{z}_n^-) = J_n^M(\bar{z}_n^+)$. Because the learning drift jumps at $\bar{z}_n$, the stationary density generically exhibits a kink there. These are the conditions we impose in the numerical solution of Appendix B.

Remark 2 (Relaxing the benchmark). Part (i) of Lemma 4 requires neither logarithmic utility nor the power form of ϕ , so the concentration of the most skilled workers in the modern sector is a robust feature of the model. Allowing $\sigma_M \neq \sigma_T$ adds the term $\frac{1}{2}(\sigma_M^2 - \sigma_T^2)[zV_n'(z) + z^2V_n''(z)]$ to the flow advantage; the bracketed expression is the second derivative of the value function with respect to $\log z$, so the term is nonnegative when $\sigma_M \geq \sigma_T$ by Lemma 2(b), and the volatility gap then reinforces the modern sector's dynamic gain, but the single-crossing argument no longer applies directly. For utility functions other than the logarithm, the proportional subsidy no longer cancels from the static gain and the affine structure behind Lemma 2(b) is lost; part (i) still applies for any strictly increasing u , since the sign of the static gain equals the sign of the wage gap. Likewise, for a general concave ϕ the elasticity of ϕ lies below one, so the static log gain remains nondecreasing and part (i) applies, but $\log \phi(e^x)$ need not be convex in $x = \log z$ and the cutoff structure of part (ii) is not guaranteed. In all these cases we verify numerically that labor supply retains the cutoff form at the estimated parameters.

²¹The constant c_π is finite for interior segmentation, $\pi^S \in (0, 1)$; under full segmentation the mixture bound in the proof is replaced by a within-sector normalization. The bound $\underline{L}$ is primitive under logarithmic utility: cross-market value differences are uniformly bounded, so the logit choice probabilities, and hence market populations, are bounded below.

A.2 Proofs

Proof of Lemma 2. (a) The increments of $\log z$ in (3) do not depend on a worker's own skill. Fix $z_1 > z_0$ and any feasible plan for a worker starting at (n, S, z_0) , that is, any state-contingent sequence of sectoral and relocation choices. Following the same plan under the same realizations of the Wiener process, relocation opportunities, and taste draws, a worker starting at z_1 obtains the skill path $z_t(z_1) = z_t(z_0)(z_1/z_0)$ and visits the same sequence of markets and sectors. Since both wage schedules are strictly increasing in z , her consumption is weakly higher at every date and strictly higher on a set of dates of positive measure. Applying this to a sequence of plans that approximate the optimum at z_0 yields $V_n^S(z_1) > V_n^S(z_0)$, and taking the upper envelope preserves the ranking. (b) Under Assumption 1, flow utility is affine in $x \equiv \log z$ in both sectors: $\log((1+b)w_n^T(z)) = \log(1+b) + \log A_n + \eta x$ and $\log((1+b)w_n^M(z)) = \log(1+b) + \log A_n - \log(1+\tau) + x$. Under any fixed plan, $x_t = x_0 + I_t$, where the increment I_t does not depend on x_0 ; hence every flow payoff is affine in x_0 with nonnegative slope along every realization, while mobility costs and taste draws enter additively. Each plan therefore delivers a discounted expected payoff that is affine in x_0 , and the value function, as the supremum of these payoffs over plans, is convex in x_0 . $\square$

Proof of Lemma 3. If either sector is empty the two peer-skill distributions in (5) coincide and the statement is trivial, so let $L_n^M, L_n^T > 0$ and write $\hat{g}_n^S \equiv N_n^S/L_n^S$ for the within-sector skill densities. By (5), each effective peer-skill distribution is a mixture of the same two components, $g_n^S = \omega^S \hat{g}_n^M + (1 - \omega^S) \hat{g}_n^T$, with weights

$$\omega^M = \frac{\pi^M L_n^M}{\pi^M L_n^M + (1 - \pi^M) L_n^T}, \quad \omega^T = \frac{(1 - \pi^T) L_n^M}{(1 - \pi^T) L_n^M + \pi^T L_n^T}.$$

Direct computation shows that $\omega^M \geq \omega^T$ if and only if $\pi^M \pi^T \geq (1 - \pi^M)(1 - \pi^T)$, which is equivalent to $\pi^M + \pi^T \geq 1$. In a sorted allocation the supports of $\hat{g}_n^T$ and $\hat{g}_n^M$ are ordered, so $\hat{g}_n^M$ first-order stochastically dominates $\hat{g}_n^T$; a mixture that places more weight on the dominating component then dominates a mixture that places less, which delivers the dominance of g_n^M over g_n^T . Finally, the vibrancy (4) is a power mean, which is monotone with respect to first-order stochastic dominance for every θ : for $\theta > 0$, z^θ is increasing, so $\int z^\theta g_n^M dz \geq \int z^\theta g_n^T dz$, and taking the power $1/\theta > 0$ preserves the ranking; for $\theta < 0$ both steps reverse, and the ranking is again preserved; the case $\theta \rightarrow 0$ follows by continuity. Strictness under the stated conditions follows from $\omega^M > \omega^T$ together with the strict dominance of the components. $\square$

Proof of Lemma 4. With $\sigma_M = \sigma_T = \sigma$, the Itô corrections in (6) cancel, so $\mu_n^M - \mu_n^T = \gamma(Z_n^M - Z_n^T)$ and the curvature term in (10) drops out:

$$\Delta_n(z) = u((1+b)w_n^M(z)) - u((1+b)w_n^T(z)) + \gamma(Z_n^M - Z_n^T)zV_n'(z).$$

(i) The wage gap $z \mapsto z/(1+\tau) - \phi(z)$ vanishes at zero, is strictly negative near zero because the Inada conditions imply $\phi(z)/z \rightarrow \infty$ as $z \rightarrow 0$, and its derivative, $1/(1+\tau) - \phi'(z)$, is strictly increasing; the gap is therefore strictly quasiconvex and crosses zero exactly once, from below, at $\hat{z}$ (it eventually turns positive since $\phi(z)/z \rightarrow 0$ as $z \rightarrow \infty$). Since u is strictly increasing, the

static gain has the same sign as the wage gap, so it is nonnegative for $z \geq \hat{z}$, while the dynamic gain is nonnegative by $Z_n^M \geq Z_n^T$ and Lemma 2(a); thus $\Delta_n(z) \geq 0$, strictly so for $z > \hat{z}$. (ii) Under Assumption 1, $u = \log$, so the subsidy cancels from the static gain and, with $\phi(z) = z^\eta$, the static gain equals $(1 - \eta) \log z - \log(1 + \tau)$. Writing $x \equiv \log z$ and $v_n(x) \equiv V_n(e^x)$, so that $z V_n'(z) = v_n'(x)$, the flow advantage becomes, with slight abuse of notation,

$$\Delta_n(x) = (1 - \eta) x - \log(1 + \tau) + \gamma (Z_n^M - Z_n^T) v_n'(x).$$

The first term is strictly increasing and unbounded in both directions. The second term is nonnegative and nondecreasing in x : $v_n' \geq 0$ by Lemma 2(a), and v_n' is nondecreasing by Lemma 2(b), which states that v_n is convex. Hence Δ_n is strictly increasing in x , with $\lim_{x \rightarrow \infty} \Delta_n(x) = \infty$; and since v_n' is nonnegative and nondecreasing, it admits a finite limit as $x \rightarrow -\infty$, so the dynamic term stays bounded while the static term diverges and $\lim_{x \rightarrow -\infty} \Delta_n(x) = -\infty$. Therefore Δ_n crosses zero exactly once, at some $\bar{x}_n \in \mathbb{R}$, and labor supply is a cutoff policy with interior cutoff $\bar{z}_n = e^{\bar{x}_n} \in (0, \infty)$. At the static crossing point $\hat{x} \equiv \log \hat{z} = \log(1 + \tau)/(1 - \eta)$, we have $\Delta_n(\hat{x}) = \gamma (Z_n^M - Z_n^T) v_n'(\hat{x}) \geq 0$, so $\bar{z}_n \leq \hat{z}$, with strict inequality whenever $Z_n^M > Z_n^T$, since $v_n' > 0$ by strict monotonicity. $\square$

Proof of Lemma 5. Fix a sorted stationary allocation whose learning drifts are bounded by γZ_H^* . First, by (5), the numerator of each effective peer-skill distribution satisfies $\pi^S N_n^S(z) + (1 - \pi^S) N_n^{-S}(z) \leq \max\{\pi^S, 1 - \pi^S\} N_n(z)$, while the denominator satisfies $\pi^S L_n^S + (1 - \pi^S) L_n^{-S} \geq \min\{\pi^S, 1 - \pi^S\} L_n$. Hence $g_n^S(z) \leq c_\pi N_n(z)/L_n \leq (c_\pi/\underline{L}) N_n(z)$, and therefore

$$(Z_n^S)^\theta = \int_0^\infty z^\theta g_n^S(z) dz \leq \frac{c_\pi}{\underline{L}} \int_0^\infty z^\theta \sum_{k=1}^N N_k(z) dz.$$

Second, by regeneration, the stationary cross-section aggregates surviving entry cohorts: a cohort of age a carries weight $\delta e^{-\delta a}$, and its θ -th moment is at most $m_\psi(\theta) e^{(\theta \gamma Z_H^* + \theta^2 \sigma^2/2)a}$, since log-skill increments have drift at most γZ_H^* and volatility at most σ . Integrating over ages,

$$\int_0^\infty z^\theta \sum_{k=1}^N N_k(z) dz \leq \frac{\delta m_\psi(\theta)}{\delta - \theta \gamma Z_H^* - \theta^2 \sigma^2/2} = \frac{(\theta + 1) \delta m_\psi(\theta)}{\delta - \theta^2 \sigma^2/2},$$

where the equality uses $\theta \gamma Z_H^* = \frac{\theta}{\theta+1} (\delta - \theta^2 \sigma^2/2)$, which also shows that the denominator is positive, so the moment is finite. Combining the two bounds, $Z_n^S \leq Z_H^*$ holds for all n and S whenever

$$\frac{c_\pi}{\underline{L}} \frac{(\theta + 1) \delta m_\psi(\theta)}{\delta - \theta^2 \sigma^2/2} \leq (Z_H^*)^\theta = \left[\frac{\delta - \theta^2 \sigma^2/2}{(\theta + 1) \gamma} \right]^\theta,$$

which rearranges exactly to (12) with $\kappa = \underline{L}/c_\pi$. Assumption 3 therefore delivers the cap. We note that the moment bound in the second step is not specific to our renewal argument: in the language of Meyn and Tweedie (1993), it is the bound $\pi(V) \leq d/c$ for the test function $V(z) = z^\theta$, whose drift condition we verify in Step 2 of the proof of Proposition 1. $\square$

Proof of Proposition 1 (sketch). We first establish existence for $\sigma_M = \sigma_T = \sigma$ (Steps 1–4) and then extend the construction to small volatility gaps (Step 5). Let Z_H^* be the vibrancy cap of Lemma 5, let $\mathcal{K} \equiv \{(Z_n^M, Z_n^T)_{n=1}^N : 0 \leq Z_n^T \leq Z_n^M \leq Z_H^*\}$, a compact convex subset of $\mathbb{R}^{2N}$, and fix $Z \in \mathcal{K}$. *Step 1.* Given the drifts implied by Z , the envelope equation (7) is a discounted, monotone system; under logarithmic utility flow payoffs are affine in $\log z$, so a standard contraction argument in a supremum norm weighted by $1 + |\log z|$ delivers a unique solution with at most logarithmic growth, jointly continuous in Z , together with the mobility shares m_{nm} and the entry probabilities Π_n . Existence and uniqueness at this step are standard: given Z , (7) is a stationary discounted problem whose nonlocal mobility term is Lipschitz and monotone, so existence follows from Perron’s method and uniqueness from the comparison principle for viscosity solutions; Achdou et al. (2022) provide an analogous treatment of coupled HJB–KFE systems in economics. Lemma 2 and Lemma 4 apply, so labor supply is a cutoff policy with thresholds $\bar{z}_n(Z) \in (0, \hat{z}]$; the strict monotonicity of the flow advantage in $\log z$ makes each threshold unique and continuous in Z . *Step 2.* Given these policies, the worker process on states (n, z) is a Feller process that regenerates at rate δ : upon death, workers restart from the entry distribution ψ combined with the logit entry choice (9). Existence and uniqueness of the invariant distribution require no PDE machinery: by regeneration, the stationary cross-section is the aggregate of surviving entry cohorts, where a cohort of age a carries weight $\delta e^{-\delta a}$ and its skill distribution is the law of the policy-induced process a periods after entry; conditioning any stationary law on the time since entry delivers the same renewal representation, which gives uniqueness. The same conclusions follow from the stability theory of continuous-time Markov processes. Because death resets every worker to the same entry law regardless of her current state, every compact set is petite in the sense of Meyn and Tweedie (1993), and the test function $V(n, z) = z^\theta$ satisfies their Foster–Lyapunov drift condition: writing $\mathcal{A}$ for the extended generator, the diffusion part contributes at most $(\theta\gamma Z_H^* + \theta^2\sigma^2/2)z^\theta$, mobility leaves V unchanged, and death-and-rebirth contributes $\delta(m_\psi(\theta) - z^\theta)$, so that

$$\mathcal{A}V \leq -cV + d, \quad c \equiv \delta - \theta\gamma Z_H^* - \frac{\theta^2\sigma^2}{2} = \frac{\delta - \theta^2\sigma^2/2}{\theta + 1} > 0, \quad d \equiv \delta m_\psi(\theta).$$

Theorems 4.2 and 4.5 of Meyn and Tweedie (1993) then deliver positive Harris recurrence, existence and uniqueness of the invariant probability, and the moment bound $\int_0^\infty z^\theta \sum_k N_k(z) dz \leq d/c$, which is the bound used in Lemma 5; and because the drift condition holds in the stronger form of their condition (CD3), their Theorem 6.1 implies that, given policies, the cross-sectional distribution converges to the invariant one exponentially fast. The invariant distribution inherits sorting from the cutoff policies, and every market is populated because the logit choice probabilities are strictly positive. *Step 3.* Let $T(Z)$ collect the vibrancies computed from this invariant distribution through (4) and (5). By Lemma 3, $T(Z)$ respects the ranking $T_n^M(Z) \geq T_n^T(Z)$, and by Lemma 5 it remains below Z_H^* , so T maps $\mathcal{K}$ into itself. Continuity of T follows from the continuity of values, thresholds, and the invariant distribution in Z , together with the uniform integrability delivered by the moment bound in Lemma 5. Brouwer’s fixed point theorem then yields $Z^* = T(Z^*)$. *Step*

4. At Z^* , compute the subsidy rate from the budget,

$$b^* = \frac{\sum_{n=1}^N \int_0^\infty \tau w_n^M(z) N_n^M(z) dz}{\sum_{n=1}^N \sum_{S \in \{M, T\}} \int_0^\infty w_n^S(z) N_n^S(z) dz},$$

which is finite and positive because first moments are finite ($\theta \geq 1$). Under logarithmic utility, the proportional rebate raises all values uniformly by $\log(1 + b^*)/(\rho + \delta)$ and alters no decision, so the allocation and b^* jointly satisfy conditions (i)–(vi) of the equilibrium definition. *Step 5 (small volatility gaps)*. A volatility gap enters the worker’s problem only through the curvature term $\frac{1}{2}(\sigma_M^2 - \sigma_T^2) z^2 V_n''(z)$ in (10) and through the Itô corrections in the drifts (6), both of which vanish continuously as $\sigma_M - \sigma_T \rightarrow 0$; interior elliptic estimates imply that the value function and its derivatives vary continuously with (σ_M, σ_T) , uniformly on compact sets. At $\sigma_M = \sigma_T$, the flow advantage is strictly increasing in $\log z$ with slope bounded below by $1 - \eta > 0$, so its unique zero is transversal; for $|\sigma_M - \sigma_T|$ small, the flow advantage therefore remains strictly increasing on the region where it can vanish, and the cutoff structure of Lemma 4 is preserved, with thresholds close to their common-volatility counterparts. Because Lemma 5 is stated with $\sigma = \max\{\sigma_M, \sigma_T\}$, the cap and the self-map property are unaffected, and Steps 2–4 go through unchanged. Hence there exists $\Sigma > 0$ such that the construction delivers a sorting equilibrium whenever $|\sigma_M - \sigma_T| < \Sigma$. Finally, suppose $\pi^M > 1/2$ and $\pi^T > 1/2$, which immediately gives $\pi^M + \pi^T > 1$. In the constructed equilibrium the cutoffs are interior and, because $\sigma > 0$, every entry cohort has full support on $(0, \infty)$, so both sectors are populated in every market; the strict version of Lemma 3 then gives $Z_n^M > Z_n^T$ for all n . In the log formulation of the proof of Lemma 4, the flow advantage at the static crossing point $\hat{x} = \log \hat{z}$ equals $\gamma(Z_n^M - Z_n^T) v_n'(\hat{x}) + \frac{1}{2}(\sigma_M^2 - \sigma_T^2) v_n''(\hat{x})$. At $\sigma_M = \sigma_T$ the second term vanishes and the first is strictly positive; with finitely many markets, the minimum of these margins is strictly positive, and all objects vary continuously with (σ_M, σ_T) along the construction. Shrinking Σ if necessary, the flow advantage therefore remains strictly positive at $\hat{z}$ in every market, which yields $\bar{z}_n < \hat{z}$ for all n . $\square$

Proof of Proposition 2 (sketch). We argue at $\sigma_M = \sigma_T$; volatility gaps below Σ are handled exactly as in Step 5 of the proof of Proposition 1. Vibrancies affect the map T through two channels, both scaled by γ : directly, through the learning drifts γZ_n^S in the KFE (8), and indirectly, through values, and hence cutoffs and mobility shares, since Z enters the worker’s problem only through the products γZ_n^S . Weighted-norm estimates on (7) bound the sensitivity of values to Z by a constant times γ ; the implicit function theorem applied to the strictly increasing flow advantage Δ_n then bounds the sensitivity of the cutoffs; and the regenerative structure bounds the sensitivity of the invariant distribution and of its moments to drifts and policies. Composing these bounds yields $\|T(Z) - T(Z')\| \leq C\gamma \|Z - Z'\|$ on $\mathcal{K}$ for a constant C that does not depend on γ , so for $\gamma < \bar{\gamma} \equiv 1/C$ the map T is a contraction and its fixed point is unique. At $\gamma = 0$ the result is immediate: learning is inert, every cutoff equals the static crossing point $\hat{z}$, and T is constant. $\square$

Proof of Lemma 1. In a stationary equilibrium the vibrancies are constant, so for a worker who remains in sector S of market n between t and $t + h$, integrating (3) yields $\log z_{t+h} - \log z_t = \gamma Z_n^S h + \sigma_S (W_{t+h} - W_t)$. Because the law of motion (3) depends only on the worker’s market

and sector, this expression holds along any path on which the worker stays in (n, S) , whether or not staying is optimal. Under Assumption 1, the wage schedules (1) are log-linear in skill, $\log w_n^M(z) = \log A_n - \log(1 + \tau) + \log z$ and $\log w_n^T(z) = \log A_n + \eta \log z$, so log-wage growth equals log-skill growth scaled by the elasticity of the wage schedule: one in the modern sector and η in the traditional sector. Taking conditional expectations eliminates the Brownian increment, so that $\mathbb{E}_t[\Delta \log w_{n,t+h}^M] = \gamma Z_n^M h$ and $\mathbb{E}_t[\Delta \log w_{n,t+h}^T] = \eta \gamma Z_n^T h$, and the difference equals $\gamma(Z_n^M - \eta Z_n^T)h$. For the sign, note that $\pi^M > 1/2$ and $\pi^T > 1/2$ imply $\pi^M + \pi^T > 1$. In a sorting equilibrium the allocation in every market is sorted at $\bar{z}_n$, and both sectors are populated because entry cohorts have full support on $(0, \infty)$; the strict version of Lemma 3 therefore yields $Z_n^M > Z_n^T$. Decomposing

$$\gamma(Z_n^M - \eta Z_n^T) = \gamma(Z_n^M - Z_n^T) + (1 - \eta) \gamma Z_n^T,$$

the first term is strictly positive because $\gamma > 0$ and $Z_n^M > Z_n^T$, and the second term is nonnegative because $\eta \in (0, 1)$ and $Z_n^T \geq 0$. Hence the expected wage-growth difference is strictly positive for every $h > 0$. $\square$

B Computational Appendix

This appendix is a self-contained summary of the equations and algorithms used to solve the model.

The numerical procedure consists of three components. First, conditional on an aggregate stationary distribution, we solve the worker's partial-equilibrium problem using a finite-difference approximation to the Hamilton–Jacobi–Bellman (HJB) equation, which yields a market-specific sector cutoff policy rule. Second, given these policy rules, we solve the Kolmogorov forward equation (KFE) for the stationary distribution, obtaining the equilibrium distribution of modern and traditional workers across markets. Third, we compute the general equilibrium by iterating jointly on the value function, the stationary distribution, and the learning drifts until the sector cutoffs converge.

B.1 HJB Numerical Solution (Partial Equilibrium)

B.1.1 Summary of Equations

Fix market productivities $\{A_n\}_{n=1}^N$, migration costs $\{\xi_{nm}\}_{n,m=1}^N$, model parameters, a transfer b , and a candidate aggregate stationary density

$$N_n(z), \quad n = 1, \dots, N.$$

In partial equilibrium the distribution is taken as given when solving the worker problem. Given sector policy weights $(\omega_{n,\text{mod}}(z), \omega_{n,\text{trad}}(z))$ with $\omega_{n,\text{mod}}(z) + \omega_{n,\text{trad}}(z) = 1$, sectoral densities are

$$N_{n,\text{mod}}(z) = \omega_{n,\text{mod}}(z) N_n(z), \quad N_{n,\text{trad}}(z) = \omega_{n,\text{trad}}(z) N_n(z). \quad (\text{B1})$$

Once the aggregate density is split across sectors, the within-sector normalized densities are

$$g_{n,s}(z) = \frac{N_{n,s}(z)}{\int N_{n,s}(x) dx}, \quad s \in \{\text{mod}, \text{trad}\}, \quad (\text{B2})$$

whenever the denominator is positive.

The sectoral vibrancy indexes, the core learning terms, are

$$\tilde{Z}_{n,s} = \left(\int z^\theta g_{n,s}(z) dz \right)^{1/\theta}, \quad s \in \{\text{mod}, \text{trad}\}, \quad (\text{B3})$$

and the corresponding drifts are

$$\mu_{n,s} = (1 - \gamma)A_n + \gamma \left[\pi_s \tilde{Z}_{n,s}^{\frac{\varepsilon-1}{\varepsilon}} + (1 - \pi_s) \tilde{Z}_{n,-s}^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}. \quad (\text{B4})$$

The wage schedules are

$$w_{n,\text{mod}}(z) = \frac{A_n z}{1 + \tau}, \quad w_{n,\text{trad}}(z) = A_n z - \varphi_n(z), \quad (\text{B5})$$

with traditional-sector cost

$$\varphi_n(z) = \lambda_\phi A_n z + \frac{e^{-\eta A_n z} - 1}{\eta}. \quad (\text{B6})$$

Given the transfer b , flow utility is

$$u_{n,s}(z) = \log(\max\{w_{n,s}(z) + b, \underline{c}\}), \quad \underline{c} > 0 \text{ small}. \quad (\text{B7})$$

The positive constant $\underline{c}$ is a numerical safeguard that avoids evaluating the logarithm at non-positive consumption levels.

Let $V_{n,s}(z)$ denote the sector-specific continuation value and let

$$V_n(z) = \max\{V_{n,\text{mod}}(z), V_{n,\text{trad}}(z)\} \quad (\text{B8})$$

be the free-switching envelope. The continuation value associated with a relocation opportunity is

$$\mathcal{C}_n[V](z) = \frac{1}{\nu} \log \left(\sum_{m=1}^N e^{\nu(V_m(z) - \xi_{nm})} \right). \quad (\text{B9})$$

For each market n , the envelope Bellman equation is

$$(\rho + \delta + \lambda_M)V_n(z) = \lambda_M \mathcal{C}_n[V](z) + \max_{s \in \{\text{mod}, \text{trad}\}} \{u_{n,s}(z) + \mathcal{B}_{n,s}V_n(z)\}, \quad (\text{B10})$$

where the drift–diffusion operator associated with learning in sector s , for a generic value function

$\phi(z)$ defined on the productivity grid, is

$$\mathcal{B}_{n,s}\phi(z) = \left(\mu_{n,s} + \frac{\sigma_s^2}{2}\right) z \partial_z \phi(z) + \frac{\sigma_s^2}{2} z^2 \partial_{zz} \phi(z). \quad (\text{B11})$$

After solving for the envelope, the sector-specific values are recovered from

$$V_{n,s}(z) = \frac{u_{n,s}(z) + \mathcal{B}_{n,s}V_n(z) + \lambda_M \mathcal{C}_n[V](z)}{\rho + \delta + \lambda_M}, \quad s \in \{\text{mod}, \text{trad}\}. \quad (\text{B12})$$

The market-level sector cutoff is then defined by the indifference condition

$$V_{n,\text{mod}}(\bar{z}_n) = V_{n,\text{trad}}(\bar{z}_n). \quad (\text{B13})$$

B.1.2 Summary of Equations For Numerical Solution

To solve the Bellman problem numerically, we bound the productivity state space by a finite interval $[z_{\min}, z_{\max}]$. Let

$$z^{\text{ex}} = \{z_j\}_{j=1}^J \subset [z_{\min}, z_{\max}], \quad z_j = z_{\min} + (j-1)\Delta z, \quad (\text{B14})$$

be the computational grid, where $\Delta z = (z_{\max} - z_{\min})/(J-1)$. In our calculations the grid is uniform, but the formulas below also clarify how the problem is written once the continuous state has been replaced by a finite collection of nodes.

The Bellman equation is approximated only on the bounded interval. The lower boundary is reflecting and the upper boundary is treated as a vanishing tail. These conditions are imposed directly in the boundary rows of the sparse matrices that approximate the drift–diffusion operator. Hence the numerical problem becomes: find values $\{V_n(z_j)\}_{n,j}$ such that, at each grid point,

$$(\rho + \delta + \lambda_M)V_n(z_j) = \lambda_M \mathcal{C}_n[V](z_j) + \max_{s \in \{\text{mod}, \text{trad}\}} \{u_{n,s}(z_j) + [\mathcal{B}_{n,s}V_n](z_j)\}, \quad (\text{B15})$$

subject to the boundary conditions embedded in the discrete operator.

Once the grid has been fixed, the remaining numerical tasks are threefold. First, one must approximate the differential operator $\mathcal{B}_{n,s}$ with a sparse matrix. Second, one must determine, node by node, which sector delivers the larger Bellman right-hand side. Third, given that sector assignment, one must solve the resulting nonlinear fixed-point problem for the envelope values.

B.1.3 HJB Discretization with Upwind Finite-Differences

Grid and discretized functions

For each market n , let

$$V_n = (V_n(z_1), \dots, V_n(z_J))' \in \mathbb{R}^J$$

denote the vector of envelope values on the grid, and analogously define the vectors of flow utilities

$$u_{n,s} = (u_{n,s}(z_1), \dots, u_{n,s}(z_J))', \quad s \in \{\text{mod}, \text{trad}\}.$$

The numerical problem is therefore to solve for the finite-dimensional object $\{V_n\}_{n=1}^N$.

Discretizing the drift–diffusion operator

For each market–sector pair (n, s) , consider a generic value function $\phi(z)$ defined on the productivity grid. The drift–diffusion operator entering the HJB is

$$\mathcal{B}_{n,s}\phi(z) = \left(\mu_{n,s} + \frac{\sigma_s^2}{2}\right) z \partial_z \phi(z) + \frac{\sigma_s^2}{2} z^2 \partial_{zz} \phi(z). \quad (\text{B16})$$

To discretize this operator on the productivity grid, we approximate the first derivative with an upwind finite difference and the second derivative with a centered second difference. This choice is standard in continuous-time problems because it preserves stability when drift dominates diffusion.

For an interior node z_j , it is convenient to define

$$a_{n,s}(z_j) \equiv \frac{\sigma_s^2}{2} z_j^2, \quad b_{n,s}(z_j) \equiv \left(\mu_{n,s} + \frac{\sigma_s^2}{2}\right) z_j,$$

so that

$$\mathcal{B}_{n,s}\phi(z_j) = a_{n,s}(z_j) \partial_{zz} \phi(z_j) + b_{n,s}(z_j) \partial_z \phi(z_j).$$

This decomposition separates the diffusion and drift components and makes clear how each enters the discrete operator.

Discrete HJB at an interior grid point

We begin with the diffusion term. Using the centered second difference,

$$\partial_{zz} \phi(z_j) \approx \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{\Delta z^2},$$

we obtain

$$a_{n,s}(z_j) \partial_{zz} \phi(z_j) \approx d_{n,s}(z_j) \phi_{j-1} - 2d_{n,s}(z_j) \phi_j + d_{n,s}(z_j) \phi_{j+1},$$

where

$$d_{n,s}(z_j) \equiv \frac{a_{n,s}(z_j)}{\Delta z^2} = \frac{\sigma_s^2 z_j^2}{2\Delta z^2}. \quad (\text{B17})$$

Next consider the drift term. The sign of the local drift determines the upwind direction. Define

$$q_{n,s}(z_j) \equiv \frac{b_{n,s}(z_j)}{\Delta z} = \left(\mu_{n,s} + \frac{\sigma_s^2}{2}\right) \frac{z_j}{\Delta z}, \quad (\text{B18})$$

and write

$$q_{n,s}^+(z_j) = \max\{q_{n,s}(z_j), 0\}, \quad q_{n,s}^-(z_j) = \max\{-q_{n,s}(z_j), 0\}.$$

If $q_{n,s}(z_j) \geq 0$, we use the forward difference

$$\partial_z \phi(z_j) \approx \frac{\phi_{j+1} - \phi_j}{\Delta z},$$

while if $q_{n,s}(z_j) < 0$, we use the backward difference

$$\partial_z \phi(z_j) \approx \frac{\phi_j - \phi_{j-1}}{\Delta z}.$$

These two cases can be written compactly as

$$b_{n,s}(z_j) \partial_z \phi(z_j) \approx q_{n,s}^-(z_j) \phi_{j-1} - (q_{n,s}^+(z_j) + q_{n,s}^-(z_j)) \phi_j + q_{n,s}^+(z_j) \phi_{j+1}.$$

Combining the diffusion and drift approximations yields the discrete HJB operator at the interior node z_j :

$$[\mathcal{B}_{n,s}\phi](z_j) \approx (d_{n,s}(z_j) + q_{n,s}^-(z_j)) \phi_{j-1} - (2d_{n,s}(z_j) + q_{n,s}^+(z_j) + q_{n,s}^-(z_j)) \phi_j + (d_{n,s}(z_j) + q_{n,s}^+(z_j)) \phi_{j+1}. \quad (\text{B19})$$

Equation (B19) makes clear that the coefficient on ϕ_j is the negative of the total local outflow, while the off-diagonal terms govern movement toward adjacent productivity nodes.

Constructing the sparse generator $A_{n,s}^{\text{HJB}}$

Equation (B19) defines the interior rows of the sparse matrix $A_{n,s}^{\text{HJB}}$ that approximates $\mathcal{B}_{n,s}$ on the productivity grid. Specifically, for an interior row j ,

$$\begin{aligned} A_{n,s}^{\text{HJB}}(j, j-1) &= d_{n,s}(z_j) + q_{n,s}^-(z_j), \\ A_{n,s}^{\text{HJB}}(j, j) &= -\left(2d_{n,s}(z_j) + q_{n,s}^+(z_j) + q_{n,s}^-(z_j)\right), \\ A_{n,s}^{\text{HJB}}(j, j+1) &= d_{n,s}(z_j) + q_{n,s}^+(z_j), \end{aligned} \quad (\text{B20})$$

with all other entries in row j equal to zero. Hence $A_{n,s}^{\text{HJB}}$ is tridiagonal in the interior and satisfies

$$\mathcal{B}_{n,s}\phi \approx A_{n,s}^{\text{HJB}}\phi.$$

This representation is useful because it translates the continuous drift-diffusion operator into a sparse linear operator that can be used directly in the HJB iteration. Once the matrices $\{A_{n,s}^{\text{HJB}}\}_{n,s}$ are assembled, they can be combined with the active-sector selection and continuation values to form the semi-implicit system solved at each iteration.

Boundary rows

The first and last rows of $A_{n,s}^{\text{HJB}}$ are modified to impose the boundary conditions described above. At the lower boundary, we impose a reflecting condition, so mass is not allowed to drift below the minimum productivity point. At the upper boundary, we impose a vanishing-tail condition, which approximates the fact that the value function becomes locally flat sufficiently far in the upper tail.

These boundary adjustments preserve the same logic as in the interior. Off-diagonal terms describe local movement across adjacent grid points, while the diagonal entry captures the total outflow implied by the local drift–diffusion approximation and the boundary condition. In this way, the full matrix $A_{n,s}^{\text{HJB}}$ provides a discrete approximation to the continuous operator on the entire bounded grid.

Lower boundary. At the first node z_1 , the process cannot move below the grid. Define

$$d_{n,s}(z_1) = \frac{\sigma_s^2 z_1^2}{2\Delta z^2}, \quad q_{n,s}^+(z_1) = \max \left\{ \left(\mu_{n,s} + \frac{\sigma_s^2}{2} \right) \frac{z_1}{\Delta z}, 0 \right\}.$$

The lower-boundary row is

$$A_{n,s}^{\text{HJB}}(1, 1) = -(d_{n,s}(z_1) + q_{n,s}^+(z_1)), \quad A_{n,s}^{\text{HJB}}(1, 2) = d_{n,s}(z_1) + q_{n,s}^+(z_1), \quad (\text{B21})$$

with all remaining entries in row 1 equal to zero. If the drift points downward at the lower bound, that outward motion is suppressed by the reflecting condition; if it points upward, the drift contribution is added to the upper-diagonal entry.

Upper boundary. At the last node z_J , define

$$d_{n,s}(z_J) = \frac{\sigma_s^2 z_J^2}{2\Delta z^2}, \quad q_{n,s}^-(z_J) = \max \left\{ - \left(\mu_{n,s} + \frac{\sigma_s^2}{2} \right) \frac{z_J}{\Delta z}, 0 \right\}.$$

With the vanishing-upper-tail condition used in the code, the upper-boundary row is

$$A_{n,s}^{\text{HJB}}(J, J-1) = 2d_{n,s}(z_J) + q_{n,s}^-(z_J), \quad A_{n,s}^{\text{HJB}}(J, J) = -(2d_{n,s}(z_J) + q_{n,s}^-(z_J)), \quad (\text{B22})$$

with all remaining entries in row J equal to zero. The factor of $2d_{n,s}(z_J)$ is the discrete counterpart of imposing a locally flat upper tail rather than reflecting mass back from an image point above the grid.

Sector comparison and cutoff recovery

Given a candidate envelope V , sectoral choice is determined by comparing the two Bellman right-hand sides evaluated on the same envelope. Define

$$D_n(z_j) \equiv [u_{n,\text{mod}}(z_j) + [A_{n,\text{mod}}^{\text{HJB}} V_n]_j] - [u_{n,\text{trad}}(z_j) + [A_{n,\text{trad}}^{\text{HJB}} V_n]_j]. \quad (\text{B23})$$

If $D_n(z_j) > 0$, the modern sector is preferred at (n, z_j) ; if $D_n(z_j) < 0$, the traditional sector is preferred. The cutoff is recovered from the first sign change of $D_n(z_j)$ along the productivity grid. If $D_n(z_j)$ and $D_n(z_{j+1})$ have opposite signs, the cutoff is obtained by linear interpolation:

$$\alpha_n = -\frac{D_n(z_j)}{D_n(z_{j+1}) - D_n(z_j)}, \quad \bar{z}_n = z_j + \alpha_n(z_{j+1} - z_j). \quad (\text{B24})$$

If no sign change is found, the cutoff is assigned to the relevant boundary, which corresponds to a corner solution in that market.

Semi-implicit fixed-point iteration

The Bellman equation is solved by pseudo-time iteration. Let V_n^k denote the envelope at iteration k , and let $\Delta_\tau > 0$ be the pseudo-time step. The migration continuation term is evaluated at V^k , and the sector assignment at each node is chosen from the sign of (B23). Let $W_{n,\text{mod}}^k$ and $W_{n,\text{trad}}^k$ be diagonal matrices collecting these node-by-node sector weights. In the present application they are effectively indicator matrices, so that each node is assigned to the sector that delivers the larger Bellman right-hand side.

Define the selected operator and flow vector by

$$A_n^{\text{sel},k} = W_{n,\text{mod}}^k A_{n,\text{mod}}^{\text{HJB}} + W_{n,\text{trad}}^k A_{n,\text{trad}}^{\text{HJB}}, \quad u_n^{\text{sel},k} = W_{n,\text{mod}}^k u_{n,\text{mod}} + W_{n,\text{trad}}^k u_{n,\text{trad}}. \quad (\text{B25})$$

Then the update for the envelope solves, market by market,

$$[(\rho + \delta + \lambda_M) + \Delta_\tau^{-1}] I_J V_n^{k+1} - A_n^{\text{sel},k} V_n^{k+1} = u_n^{\text{sel},k} + \lambda_M \mathcal{C}_n[V^k] + \Delta_\tau^{-1} V_n^k. \quad (\text{B26})$$

This step treats the selected drift–diffusion operator implicitly and the nonlinear terms explicitly. The resulting linear system is sparse and can therefore be solved efficiently for each market.

The iteration is initialized either from the previous outer-loop value function or from a static approximation,

$$V_n^0(z_j) = \frac{\max\{u_{n,\text{mod}}(z_j), u_{n,\text{trad}}(z_j)\}}{\rho + \delta + \max\{\lambda_M, \varepsilon_0\}}, \quad \varepsilon_0 > 0 \text{ small}. \quad (\text{B27})$$

Iteration continues until the sup norm of successive envelope values falls below the prescribed tolerance. Once the envelope has converged, sector-specific values are recovered from (B12), and the cutoff is obtained from (B24).

B.2 KFE Numerical Solution (Partial Equilibrium)

B.2.1 Summary of Equations

Once the value function has been computed, workers switch freely between sector states according to the policy implied by the Bellman comparison. It is therefore convenient to solve first for the

aggregate stationary density $N_n(z)$ and then recover sectoral densities from the policy. Let

$$\pi_n^0(z) = \frac{e^{\nu V_n(z)}}{\sum_{m=1}^N e^{\nu V_m(z)}} \quad (\text{B28})$$

denote the probability that a newborn worker selects market n at productivity z , and let $\psi(z)$ denote the exogenous newborn skill density.

Given a sector assignment, define the active forward operator

$$\mathcal{A}_n^{\text{act},*} = W_{n,\text{mod}} A_{n,\text{mod}}^{\text{KFE}} + W_{n,\text{trad}} A_{n,\text{trad}}^{\text{KFE}}, \quad A_{n,s}^{\text{KFE}} = (A_{n,s}^{\text{HJB}})', \quad (\text{B29})$$

where the diagonal matrices $W_{n,s}$ include the sectoral policy weights:

$$W_{n,s} = \text{diag}(\omega_{n,s}(z_1), \dots, \omega_{n,s}(z_J)). \quad (\text{B30})$$

Thus the forward operator is the discrete adjoint of the HJB generator. At each productivity level z , migration across markets is governed by the multinomial logit shares

$$m_{nm}(V(z)) = \frac{e^{\nu(V_m(z) - \xi_{nm})}}{\sum_{\ell=1}^N e^{\nu(V_\ell(z) - \xi_{n\ell})}}. \quad (\text{B31})$$

The stationary KFE for the aggregate density is

$$0 = \mathcal{A}_n^{\text{act},*} N_n(z) + \lambda_M \left(\sum_{k=1}^N m_{kn}(V(z)) N_k(z) - N_n(z) \right) - \delta N_n(z) + \delta \psi(z) \pi_n^0(z). \quad (\text{B32})$$

Equation (B32) makes clear that the density evolves through four channels: learning-induced drift and diffusion in productivity, migration across markets, death, and newborn entry.

After solving for the aggregate density, sectoral densities are recovered from the policy weights:

$$N_{n,\text{mod}}(z) = \omega_{n,\text{mod}}(z) N_n(z), \quad N_{n,\text{trad}}(z) = \omega_{n,\text{trad}}(z) N_n(z). \quad (\text{B33})$$

B.2.2 Summary of Equations For Numerical Solution

As in the HJB problem, the KFE is solved on the bounded grid $\{z_j\}_{j=1}^J$. Let

$$N_n = (N_n(z_1), \dots, N_n(z_J))' \in \mathbb{R}^J$$

denote the vector of aggregate density values in market n . The numerical problem is therefore to find $\{N_n\}_{n=1}^N$ satisfying the discretized counterpart of (B32) on the finite grid.

The forward equation inherits the same treatment of boundaries used in the Bellman problem. Because the drift–diffusion operator in the KFE is the adjoint of the HJB operator, the corresponding boundary conditions are built into the first and last rows of the transposed sparse matrices. In ad-

dition, after solving the linear system, the resulting density is renormalized to satisfy the unit-mass condition

$$\sum_{n=1}^N \int N_n(z) dz = 1. \quad (\text{B34})$$

Hence the numerical KFE problem becomes a finite-dimensional sparse linear system coupled across markets through migration and newborn inflows.

B.2.3 KFE Discretization

Discrete forward operator

The discretization starts from the same sparse operators used in the HJB problem. For each market and sector, the forward operator is the transpose of the corresponding HJB matrix:

$$A_{n,s}^{\text{KFE}} = (A_{n,s}^{\text{HJB}})'$$

This is the standard relationship between backward and forward equations, given the uniform productivity grid. Once the sector assignment has been determined, the selected forward operator is assembled according to (B29). As a result, within each market the density evolves under the law of motion implied by the sector chosen at each productivity node.

Stacked system across markets

Let $\mathbf{n} \in \mathbb{R}^{NJ}$ stack the densities $\{N_n(z_j)\}_{n=1,\dots,N}^{j=1,\dots,J}$. The stationary KFE (B32) can then be written as a global sparse linear system

$$G\mathbf{n} + \mathbf{b} = 0, \quad (\text{B35})$$

where the operator G can be decomposed into two components. First, each market block contains the within-market forward operator $\mathcal{A}_n^{\text{act},*}$ together with the death term $-\delta I_J$. Second, for each productivity node, markets are coupled through the migration matrix

$$\lambda_M(m(V(z_j))' - I_N), \quad (\text{B36})$$

where $m(V(z_j))$ collects the shares in (B31). The vector $\mathbf{b}$ collects newborn inflows of the form $\delta\psi(z_j)\pi_n^0(z_j)$.

Writing the problem in this stacked form is useful because all interactions—drift, diffusion, migration, death, and entry—are represented in a single sparse linear system. Equivalently, one may write

$$(-G)\mathbf{n} = \mathbf{b}. \quad (\text{B37})$$

This system is solved directly using sparse linear-algebra methods.

Post-processing and recovery of sectoral densities

After solving (B37), the vector solution is reshaped into an $N \times J$ array. Small negative entries generated by numerical error are set to zero, and the distribution is renormalized so that (B34) holds exactly on the grid. The sectoral distributions are then reconstructed from (B33). These distributions are the objects that enter the updating formulas for vibrancy, drifts, and the transfer in general equilibrium.

B.3 General-Equilibrium Solution Algorithm

B.3.1 Summary of Equations

General equilibrium requires solving jointly for the aggregate stationary density $\{N_n(z)\}_{n=1}^N$, the free-switching value function $\{V_n(z)\}_{n=1}^N$, and the cutoff vector $\{\bar{z}_n\}_{n=1}^N$. It also requires the learning drifts $\{\mu_{n,\text{mod}}, \mu_{n,\text{trad}}\}_{n=1}^N$ and the transfer b . The fixed-point problem combines the Bellman equation (B10), the stationary forward equation (B32), the learning rule (B4), and the balanced-budget transfer condition

$$b = \tau \sum_{n=1}^N \int N_{n,\text{mod}}(z) w_{n,\text{mod}}(z) dz. \quad (\text{B38})$$

An equilibrium is therefore a collection of objects such that: (i) workers optimize given the stationary distribution and transfer; (ii) the stationary distribution is consistent with optimal behavior; (iii) learning drifts are induced by the resulting sectoral distributions; and (iv) the transfer is financed by modern-sector labor tax revenues.

B.3.2 Summary of Equations For Numerical Solution

The numerical equilibrium problem is a fixed point in distributions, values, cutoffs, drifts, and the transfer. It is convenient to use the cutoff vector as the main convergence object because the sectoral allocation enters both the Bellman and the forward equations through the node-by-node sector assignment.

The outer iteration starts from an initial density that is Pareto in productivity and assigns positive mass to every market. Specifically, let

$$g_0(z) = \frac{\theta_p \kappa^{\theta_p}}{z^{\theta_p+1}}, \quad z \geq \kappa, \quad (\text{B39})$$

be the initial productivity density, projected onto the computational grid and renormalized. Combining this with market weights $\{\omega_n^0\}_{n=1}^N$ yields the initial market distributions

$$N_n^0(z) = \omega_n^0 g_0(z), \quad \sum_{n=1}^N \int N_n^0(z) dz = 1. \quad (\text{B40})$$

The initial cutoff in each market is set equal to the mean productivity under N_n^0 , and the transfer starts from an initial scalar guess.

Given a current iterate $(N^r, \bar{z}^r, b^r)$, one computes the implied sectoral distributions, updates the learning drifts from (B4), solves the HJB, solves the stationary KFE, updates the transfer from (B38), and checks whether the cutoff vector has converged. This yields a finite-dimensional fixed-point iteration over the equilibrium objects.

B.3.3 General Equilibrium Solution Algorithm

Let superscript r denote the general-equilibrium iteration.

- Step 1. Split the current distribution across sectors.** Given $(N^r, \bar{z}^r)$, construct the policy weights $(\omega_{n,\text{mod}}^r, \omega_{n,\text{trad}}^r)$ and the sectoral densities using (B1).
- Step 2. Update learning objects.** Compute the within-sector densities (B2), the vibrancy indexes (B3), and the drift terms (B4).
- Step 3. Update wages and the transfer.** Compute the wage schedules from (B5) and update the transfer using the balanced-budget rule (B38).
- Step 4. Solve the worker problem.** Using the current drifts and transfer, solve the envelope Bellman equation by the finite-difference procedure described in Section A. This yields updated value functions and a new cutoff vector $\bar{z}^{r+1}$.
- Step 5. Solve for the stationary distribution.** Holding the updated policy fixed, solve the stationary KFE by the method described in Section B. This yields an updated aggregate distribution N^{r+1} and, through (B33), updated sectoral densities.
- Step 6. Check convergence.** Compute

$$\max_n |\bar{z}_n^{r+1} - \bar{z}_n^r|.$$

If this distance is below the prescribed tolerance, stop. Otherwise set $r \leftarrow r + 1$ and return to Step 1.

The output of this outer loop is a stationary equilibrium in which the value function, the sectoral allocation, the stationary distribution, the drift terms, and the transfer are mutually consistent.

C Additional Figures

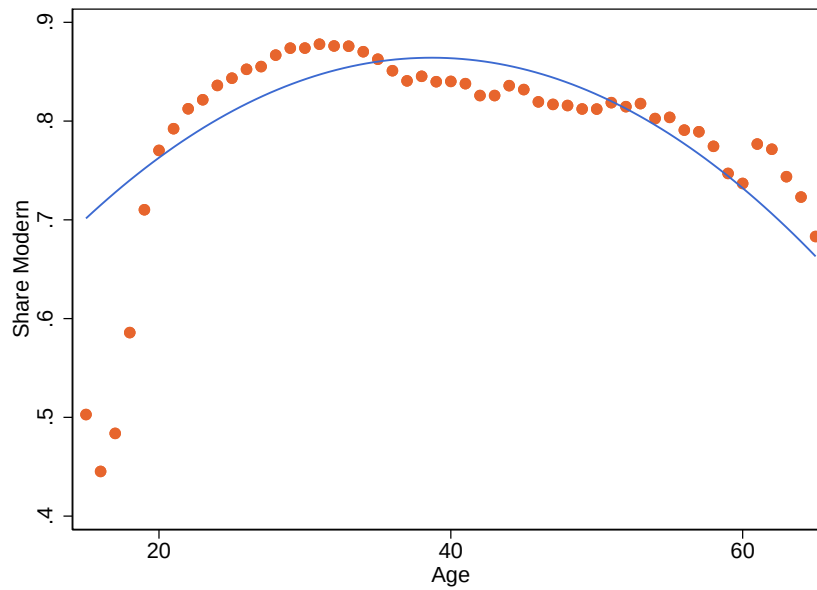
C.1 Descriptive Statistics

This section reports additional descriptive statistics on the share of modern workers in Chile. We confirm previously well-established patterns for the Chilean economy:²²

1. The share of modern workers follows an inverse U-shaped pattern over the life cycle.
2. The share of modern workers increases with educational attainment.
3. The share of modern workers increases with firm size.

Then, we report the share of modern workers by occupation and economic sector.

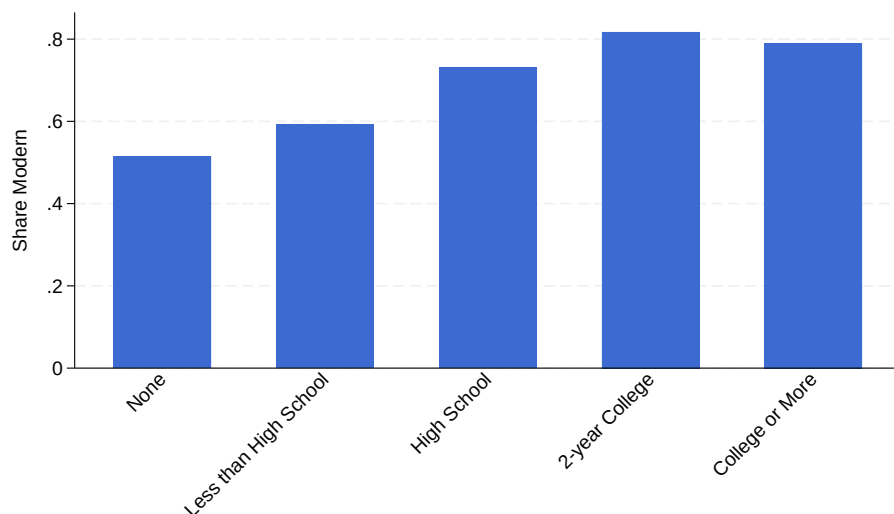
Figure C1: Share of Modern Workers by Age



Notes: Figure C1 displays the average share of modern workers at each age, together with a quadratic fit. We restrict our sample to salaried workers, excluding self-employed workers, entrepreneurs, workers without remuneration, and public-sector and armed forces employees.

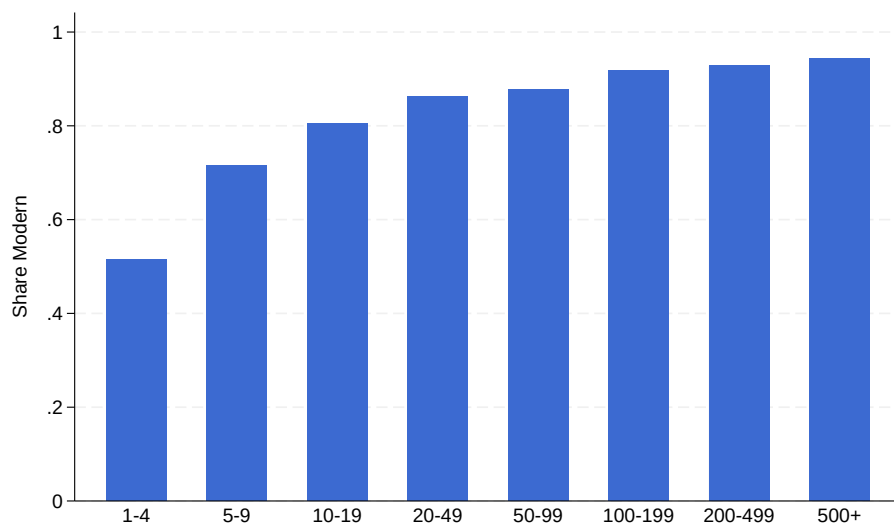
²²See Ulyssea (2020a) for a comprehensive review of informal workers empirical regularities.

Figure C2: Share of Modern Workers by Educational Attainment



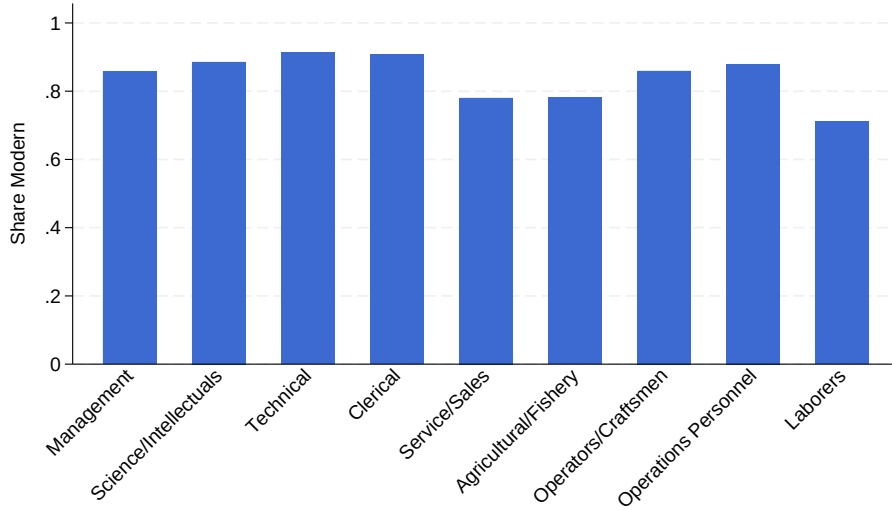
Notes: Figure C2 displays the share of modern workers by education group. We consider all workers in our baseline sample, including self-employed workers and entrepreneurs.

Figure C3: Share of Modern Workers by Firm Size



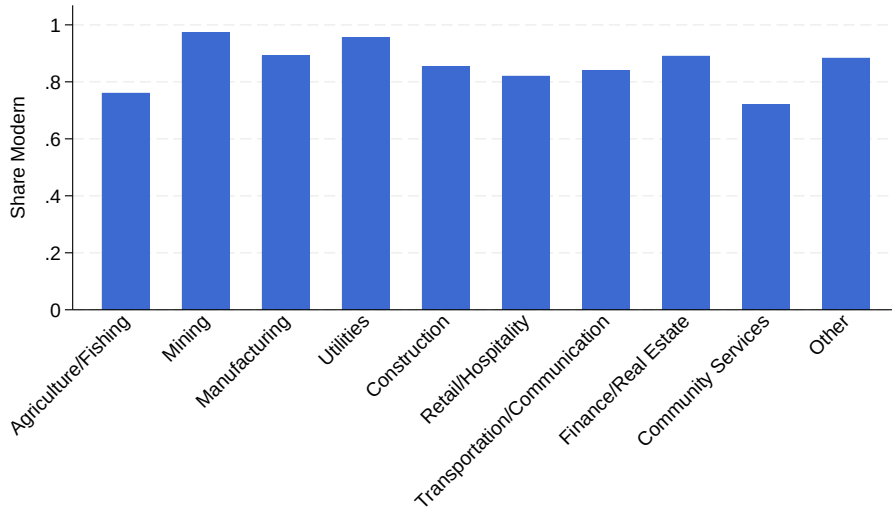
Notes: Figure C3 displays the share of modern workers by firm size. Bars denote firm size groups, and labels denote the range of employees in each group. We restrict our sample to salaried workers, excluding self-employed workers, entrepreneurs, workers without remuneration, and public-sector and armed forces employees.

Figure C4: Share of Modern Workers by Occupation



Notes: Figure C4 displays the share of modern workers by occupation. We restrict our sample to salaried workers, excluding self-employed workers, entrepreneurs, workers without remuneration, and public-sector and armed forces employees.

Figure C5: Share of Modern Workers by Sector

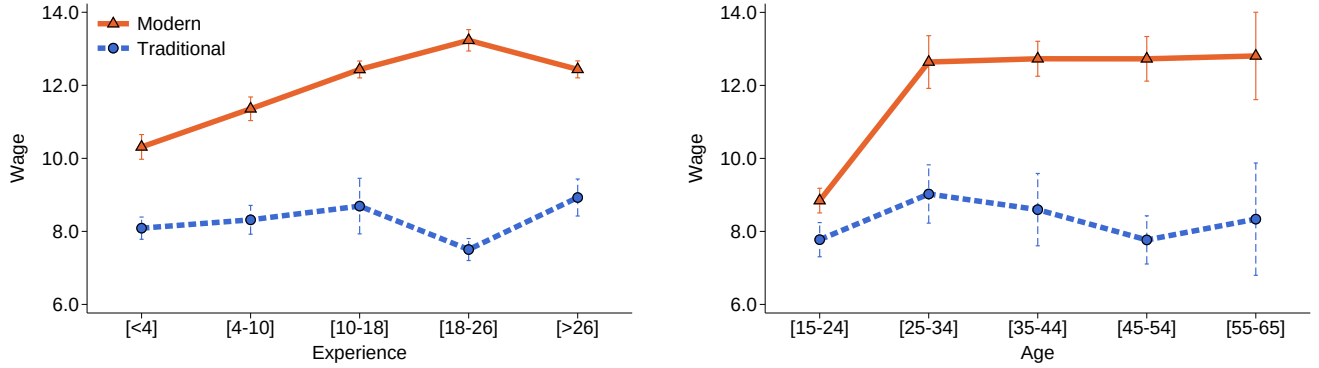


Notes: Figure C5 displays the share of modern workers by economic sector. We restrict our sample to salaried workers, excluding self-employed workers, entrepreneurs, workers without remuneration, and public-sector and armed forces employees.

C.2 Modern and Traditional Wage Dynamics

In this section, we report wage-experience and wage-age profiles for modern and traditional workers, restricting the sample to salaried workers.

Figure C6: Wages Over the Life Cycle for Modern and Traditional Workers, Salaried Workers
(a) Wage-experience profiles **(b) Wage-age profiles**



Notes: Figure C6 displays average wages by experience quintile (panel (a)) and by 10-year age bin (panel (b)) for modern and traditional workers. We restrict our sample to salaried workers, excluding self-employed workers and entrepreneurs. Vertical dashed lines denote 95 percent confidence intervals.

D Additional Tables

Table D1 reports estimates of the modern-traditional wage gap from equation (13), discussed in Section 3.

Table D1: Modern-Traditional Wage Gap

	Dependent variable: $\log w_{i,t}$		
	(1)	(2)	(3)
Modern	0.395*** (0.015)	0.093*** (0.014)	0.082*** (0.018)
Corr(Modern _{it} , α_i)			0.095 (0.011)
Observations	58,714	58,714	58,714
Adj R-squared	0.074	0.459	0.841
Controls		✓	✓
Worker Fixed Effects			✓

Notes: Table D1 reports estimates for β in equation (13). Column (1) displays the raw gap without any controls. Column (2) controls for worker characteristics, including experience in the modern and traditional sectors and firm size fixed effects. Column (3) adds worker fixed effects. Controls include age, education, gender, occupation, industry, region, firm size, experience in each sector, and time fixed effects. Corr(Modern_{it}, α_i) reports the correlation between the modern-sector indicator and the estimated worker fixed effects across worker-year observations, weighted by survey expansion factors, with standard errors clustered at the worker level. Standard errors in parentheses are clustered at the worker level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D2 reports estimates from a pooled specification that allows the peer effect in equation (16) to differ for workers above and below the mean wage of their labor market.

Table D2: Peer Effects on Wage Growth: Workers Above and Below Average Wage

	Dependent variable: $\Delta \log w_{i,t+h}$				
	Time horizon h in years				
	1	2	3	4	5
Peer wages $\times$ Above	0.029*** (0.010)	0.081*** (0.014)	0.116*** (0.019)	0.144*** (0.026)	0.109*** (0.036)
Peer wages $\times$ Below	0.014* (0.009)	0.036*** (0.013)	0.040*** (0.015)	0.069*** (0.019)	0.074*** (0.024)
Observations	39,338	29,997	21,577	15,124	9,704
Adj R-squared	0.086	0.163	0.224	0.267	0.266

Notes: Table D2 reports estimates of a pooled version of equation (15) in which log peer wages are interacted with indicators for the worker's wage being above or below the leave-one-out mean wage of her labor market, and the above-mean indicator enters directly (coefficient not reported). Controls are the same as in Figure 4, with coefficients common to the two groups. Standard errors in parentheses are clustered at the labor-market level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Quantitative Appendix

This appendix describes how the solved stationary equilibrium is converted into a synthetic worker panel for estimation. The purpose of the simulation is to translate the equilibrium objects from the computational appendix into annual worker histories that can be summarized in the same moments observed in the data.

E.1 Simulation of Worker Panel

The simulation starts from the stationary equilibrium characterized in the computational appendix. In particular, it uses four objects from the solved model:

1. The stationary distribution over market and productivity states, denoted by $N_n(z)$ in continuous time and by the stacked vector $\mathbf{n} \in \mathbb{R}^{NJ}$ on the discretized grid.
2. The policy rule $(\omega_{n,\text{mod}}(z), \omega_{n,\text{trad}}(z))$, which determines the probability of modern and traditional work at each state.
3. The associated wage schedules $(w_{n,\text{mod}}(z), w_{n,\text{trad}}(z))$.
4. The continuous-time law of motion for the joint market–productivity process, summarized by the generator matrix Q .

The last object is the key input for the simulation. Let Q denote the continuous-time generator for the discretized joint state (n, z) , where n indexes markets and z indexes productivity on the grid. Thus, Q is an $NJ \times NJ$ matrix that summarizes the same state transitions used in the stationary KFE. Within each market it contains productivity drift and diffusion, and across markets it contains migration flows governed by the value-based migration rule.²³ In this sense, the simulation uses exactly the same law of motion as the stationary distribution problem rather than introducing a separate transition approximation.

To simulate the model at a monthly frequency, we convert this continuous-time generator into a one-step transition matrix,

$$\Pi(\Delta t) = \exp(\Delta t Q), \quad \Delta t = 1/12.$$

This matrix gives the probabilities of moving from one discretized (n, z) state to another over one month.

A synthetic cross section of workers is then initialized by drawing states from the stationary distribution implied by the solved equilibrium. The code also assigns each worker an initial level of

²³Formally, for each market n , Q reproduces the first two terms of the stationary KFE in the computational appendix,

$$\mathcal{A}_n^{\text{act},*} N_n(z) + \lambda_M \left(\sum_{k=1}^N m_{kn}(V(z)) N_k(z) - N_n(z) \right).$$

Deaths and newborn inflows are added separately in the simulation.

accumulated experience so that the starting cross section is close to the stationary age profile generated by the death process. Conditional on a worker’s current (n, z) state, modern or traditional sector status is drawn using the model-implied policy weights. Wages are then assigned from the corresponding equilibrium wage schedule.

Each simulated month has three steps. First, given the worker’s current state, the simulation draws whether the worker is in the modern or traditional sector during that month and updates total, modern, and traditional experience accordingly. Second, the worker moves across market–productivity states according to the transition matrix $\Pi(1/12)$. Third, the worker faces the exogenous death hazard from the model. When a death occurs, that worker is replaced by a newborn entrant drawn from the model-implied entry distribution over markets and productivity. Newborn workers start with zero experience.

It is useful to separate the roles of these two mechanisms. The matrix Q governs movement across the discretized market–productivity state space, while death and replacement are imposed separately in the simulation because the code tracks individual worker histories. This is why turnover is handled outside the matrix transition step even though the stationary distribution itself is consistent with the same birth–death structure.

At the end of each simulated year, the code records one observation for every worker. The resulting annual panel contains worker identifiers, calendar year, market, sector status, wage, log wage, and the worker’s accumulated total, modern, and traditional experience. In the baseline implementation, the panel contains 2,500 workers simulated for 15 years.

E.2 Construction of Model Moments

The annual simulated panel is then used to construct the moments matched in estimation.

The first set of moments is the wage-transition matrix. In each simulated year, workers are assigned to wage quartiles using that year’s cross-sectional wage distribution. The code then links observations in year t and year $t + 1$ by worker identifier and computes the implied 4×4 transition matrix across quartiles. Averaging these matrices across adjacent years yields the model counterpart to the empirical wage-transition moments.

The second set of moments is the modern–traditional transition matrix. Using the same year-to-year matching procedure, the code computes the 2×2 transition matrix between modern and traditional employment and averages it across adjacent years. Because deceased workers are replaced with new identifiers, these transitions are constructed only from workers who are observed in consecutive annual snapshots.

The remaining moments are cross-sectional summaries of the simulated equilibrium. The traditional-sector employment share is the average share of worker-year observations in traditional employment. The modern-sector wage premium is computed directly from the solved wage schedules as the stationary-mass-weighted average of the ratio of modern to traditional wages across states. Finally, the model’s young-worker wage-dispersion moment is the variance of log wages among workers with at most two years of total experience in the simulated annual panel.